

Self-Supervised Learning for Robot Perception

INF8422: Robot Perception and Spatial Intelligence

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**POLYTECHNIQUE
MONTREAL**



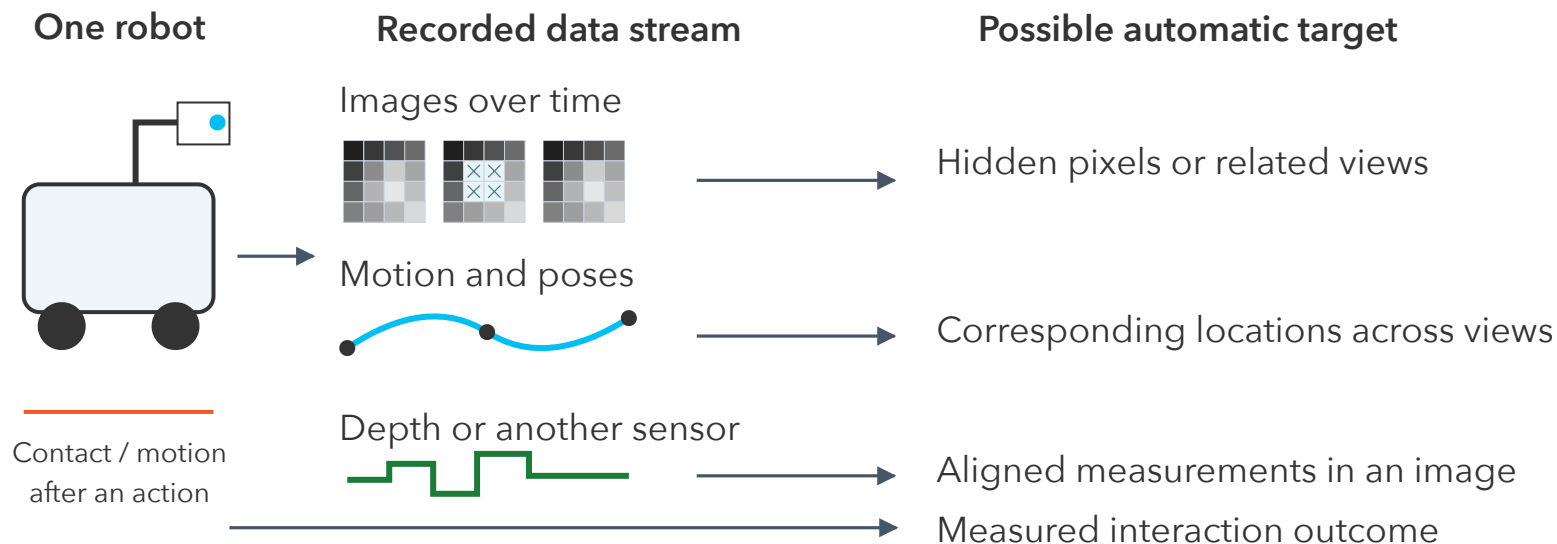
Camera observation and estimated depth
[Godard et al., Monodepth2, ICCV 2019, Fig. 1 \(top row\)](#)

SECTION 01

Self-supervised learning mechanics



Supervision available to a robot

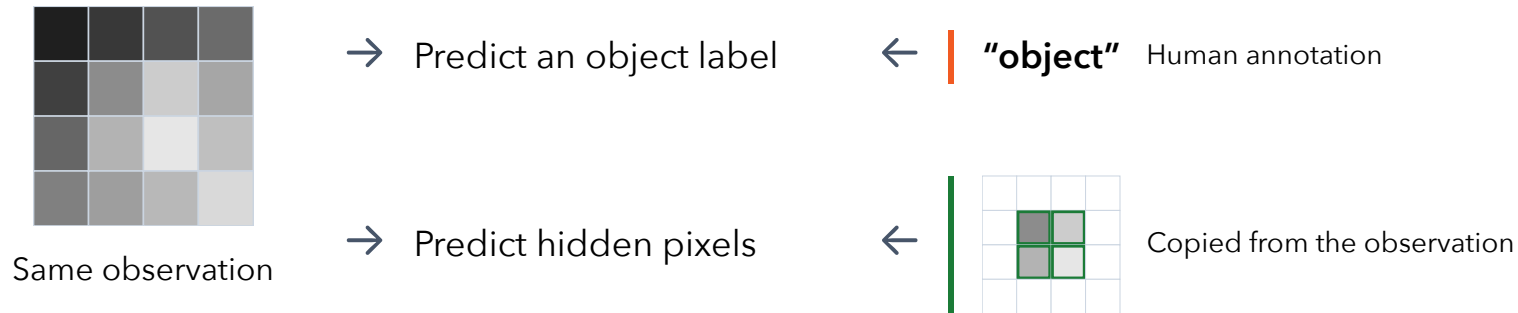


The target needs a construction rule

Recorded data can supply a training signal when we define a valid relationship between observations.

Definition of self-supervised learning

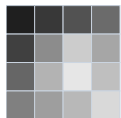
Self-supervised learning trains a model using supervisory signals derived automatically from the data itself, without requiring manually annotated labels for that learning stage.



The origin of the training target determines the supervision setting.

Supervised, self-supervised, and semi-supervised learning

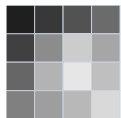
Supervised



Manually labeled examples

→ Annotation-based loss

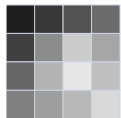
Self-supervised



Automatically constructed targets

→ Data-derived loss

Semi-supervised



Some labeled + unlabeled examples

→ Supervised + unlabeled-data terms

The loss (cost) function alone does not identify the setting

Squared error and cross-entropy can compare predictions to either manual or automatic targets.

Pretext tasks and automatic targets



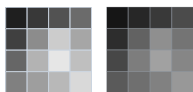
Hidden-pixel prediction

Input: visible information



Withheld observed pixels

Target: selected from the original image



Related-view matching

Input: two images with a known relationship



A known pair association

Target: known relation (e.g., rotation, translation, overlap)



Direct outcome prediction

Input: an observation before the action

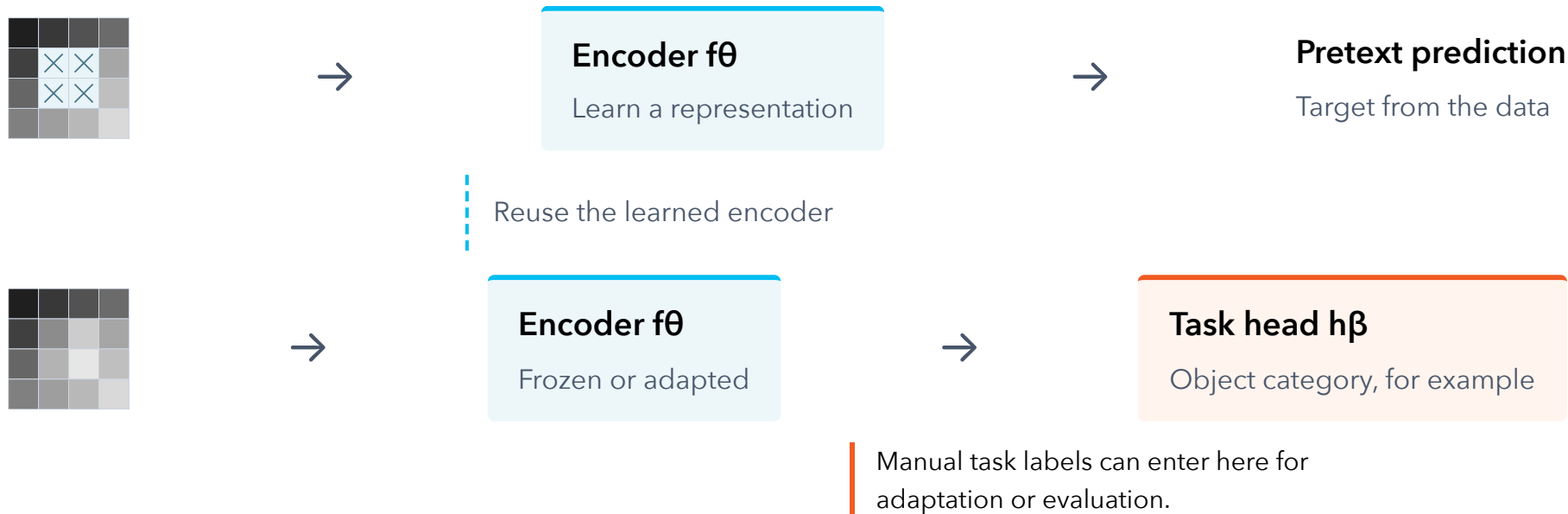


A measured traversal cost: 0.4

Target: measurements after the action

A pretext task builds a representation through prediction or matching.

Pretraining and optional fine-tuning



Two adaptation choices

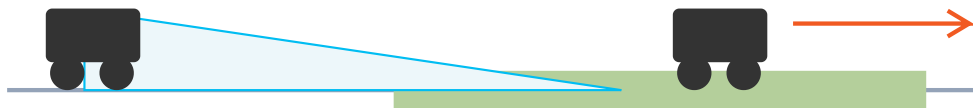
Keep the encoder fixed and train a task head, or fine-tune the encoder together with the head.

Direct learning from measured outcomes

See the terrain

Execute a known fixed motion

Measure the outcome



Commanded speed: 1.0 m/s
Measured speed: 0.6 m/s

Automatic target Illustrative cost = $1 - \text{measured} / \text{commanded} = 0.4$

Earlier camera image

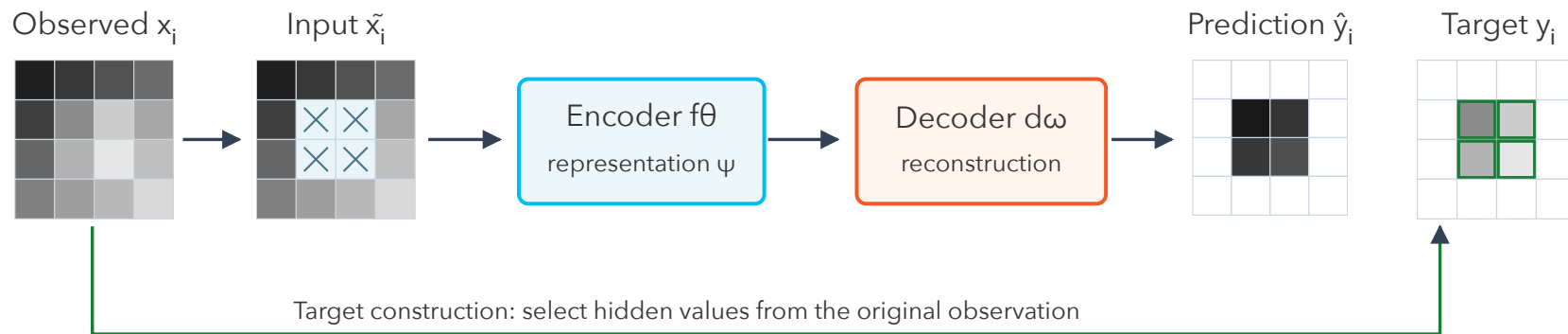


Encoder + task head



Predicted traversal cost

Masking: Input, prediction, and target

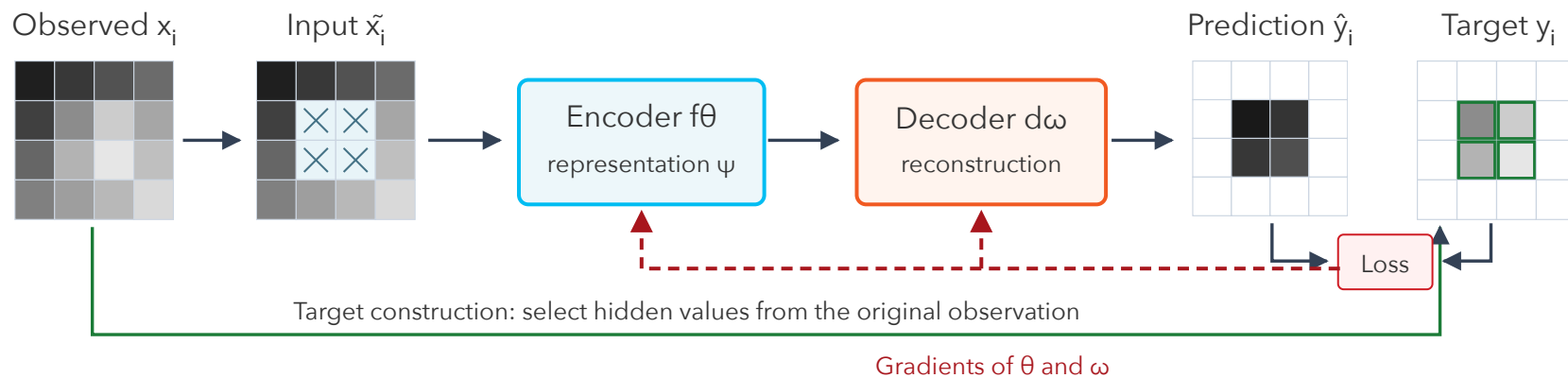


$$y_i = (x_{i,k})_{k \in \mathcal{M}}$$

Copy the withheld observed values into the target branch.

i identifies an observation · Θ contains the encoder and decoder parameters · $\mathcal{M}$ selects hidden locations

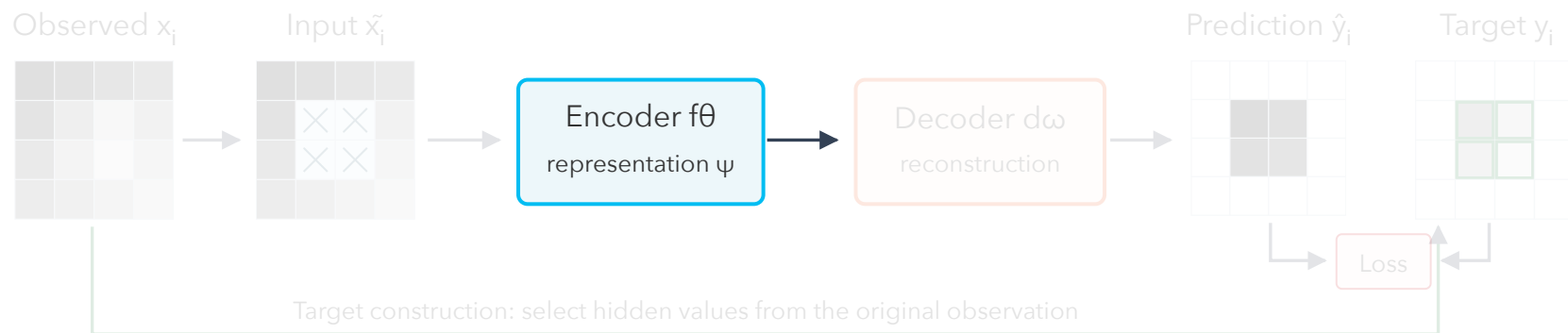
Masking: Loss and parameter update



$$\Theta_{n+1} = \Theta_n - \eta \nabla_{\Theta} \mathcal{L}$$

n counts optimization steps. η controls the size of each parameter update.

Masking: What remains after training



Keep f_θ · replace d_w with a task head when appropriate

$$\psi = f_\theta(x) \quad \hat{y}^{\text{task}} = h_\beta(\psi)$$

Use the encoder on a new observation. The task determines the output head.



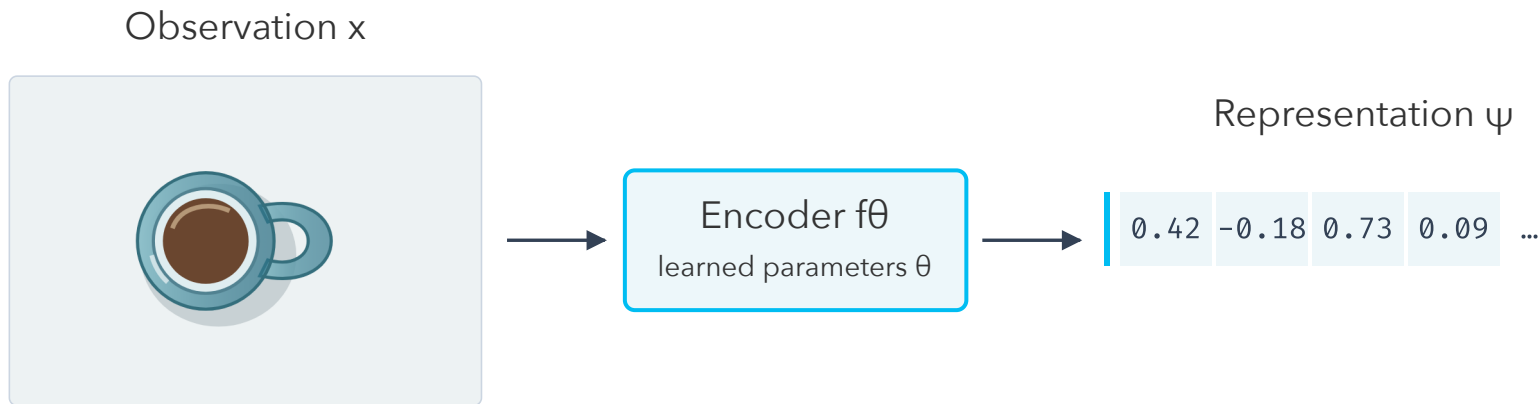
Evaluate the retained representation on the intended perception task.

SECTION 02

Representations for perception



Encoder representations



An internal description that a task can reuse

$$\psi = f_\theta(x)$$

A reusable description

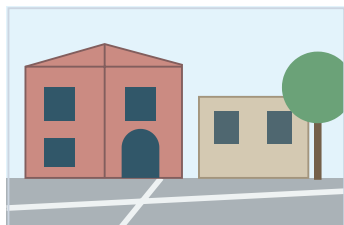
Representation quality depends on the information available for the intended task.

Global and dense features

Global features: one vector per image

Recognize a previously visited place

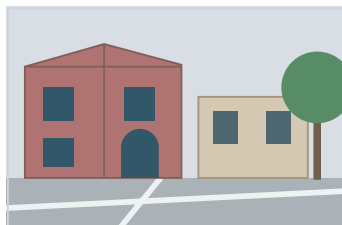
Query image



Encoder $f_\theta \downarrow$

0.42 -0.18 0.73

Stored image



Encoder $f_\theta \downarrow$

0.40 -0.20 0.71

Compare whole-image descriptors

Dense features: vectors at spatial locations

Find the corresponding handle point

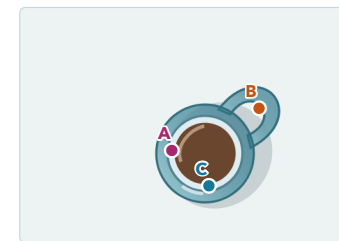
Reference image



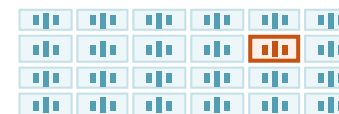
Encoder $f_\theta \downarrow$



New image



Encoder $f_\theta \downarrow$



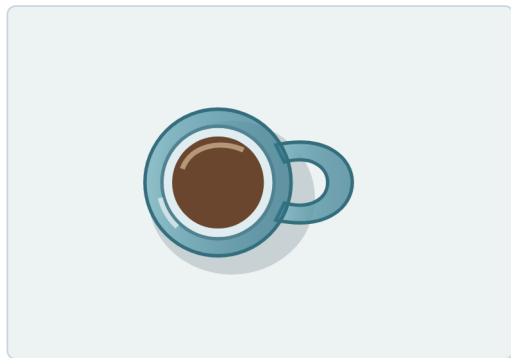
-0.24 0.81 0.46

Invariance to irrelevant changes

For **recognizing this mug**, lighting and background can be nuisance variables.

We therefore use **augmentations** during training.

Reference observation



→
change
appearance

Changed observation



Desired identity output

same mug

Lighting: 0.55×

Background: pattern

Pixels change; the recognition decision stays stable.

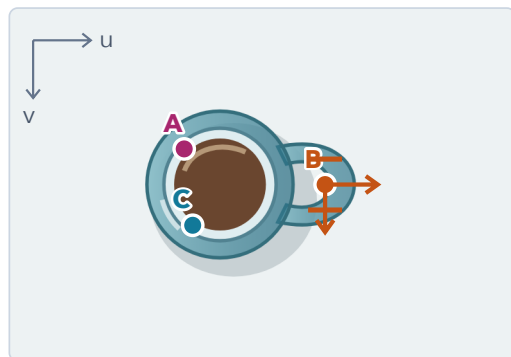
Keep the information that identifies the object across the chosen changes.

Spatial information needed for action

For **grasping the handle**, its location and orientation must remain accessible.

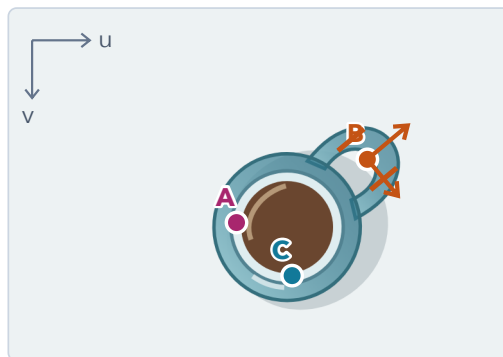
We therefore use **augmentations** during training.

Reference observation



→
move the mug

Changed observation



Grasp in image coordinates

(212.5, 90.1)

$\alpha = -40^\circ$ · positive clockwise

B: same illustrative descriptor

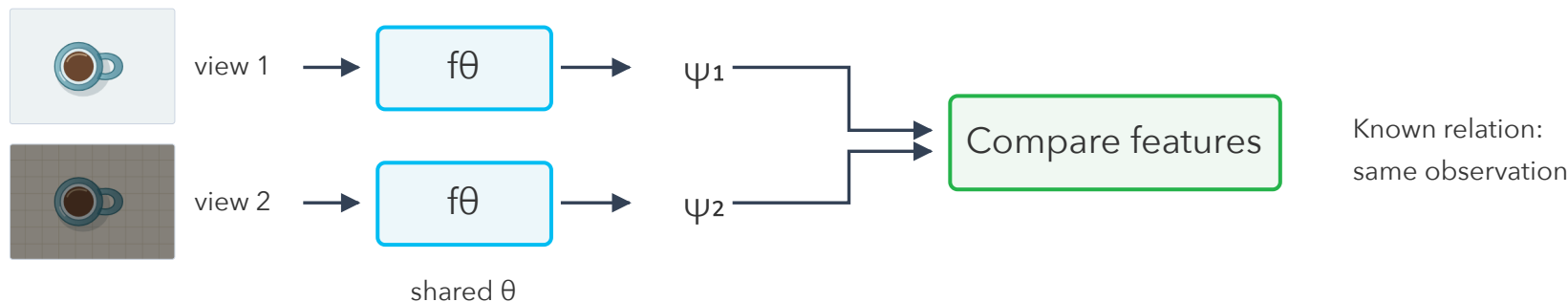
-0.24	0.81	0.46
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The handle descriptor stays associated with B; its image location and grasp orientation change.

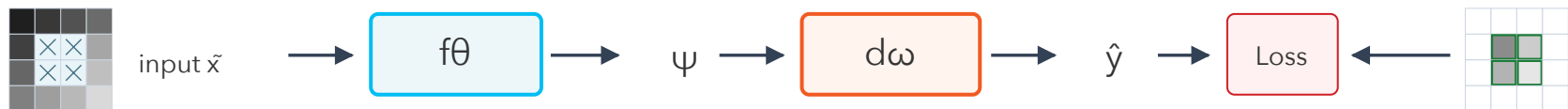
Corresponding features can stay consistent while their spatial locations change.

Matching and prediction objectives

Match related observations



Predict hidden information



Two mechanisms to follow

Matching uses a known relation between observations.

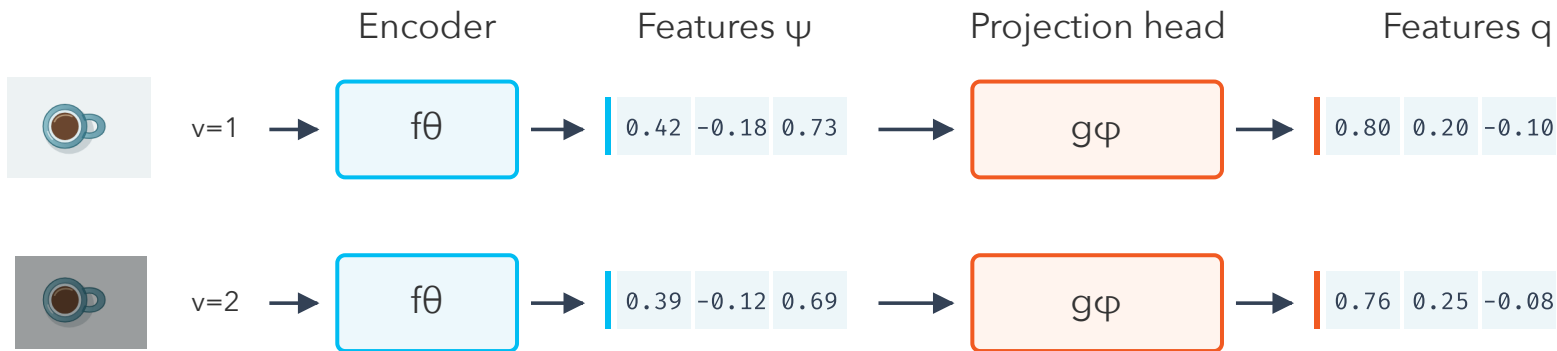
Masking uses observed information withheld from the input.

SECTION 03

Matching, collapse, and self-distillation



Encoder and projection head



Reuse the encoder representation

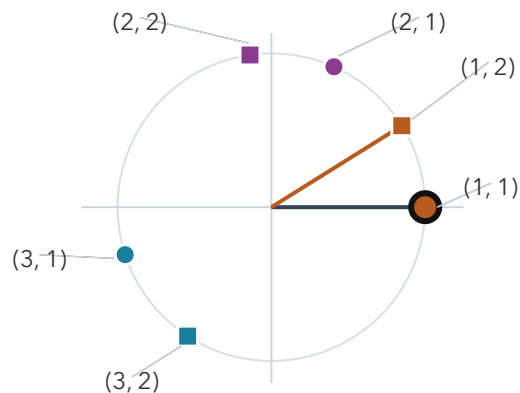
Compare projected features

$$\psi_i^{(v)} = f_\theta(\tilde{x}_i^{(v)}) \quad \mathbf{q}_i^{(v)} = g_\phi(\psi_i^{(v)})$$

The objective compares q . Many different downstream tasks can reuse encoder features ψ .

Similarity and temperature

Anchor (1, 1) · Positive (1, 2)



Candidate (i, v)

(1, 2) · positive
(2, 1) · negative
(2, 2) · negative
(3, 1) · negative
(3, 2) · negative

cosine

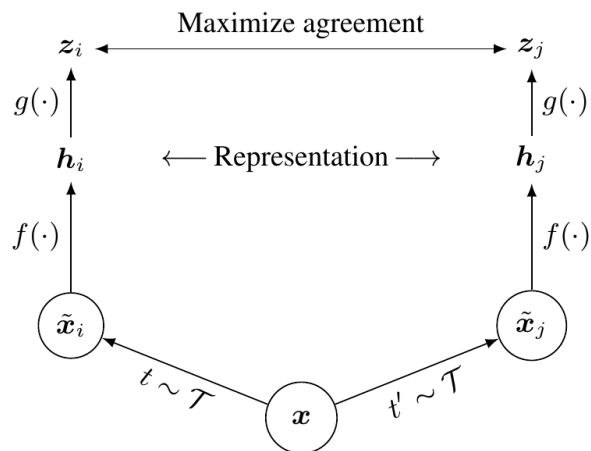
Normalized probability

0.848		60.9%
0.407		25.2%
-0.139		8.5%
-0.951		1.7%
-0.545		3.8%

$\tau = 0.50 \cdot \text{positive probability} = 60.9\% \cdot \ell = 0.496$

$$\text{sim}(\mathbf{q}_a, \mathbf{q}_b) = \frac{\mathbf{q}_a^T \mathbf{q}_b}{\|\mathbf{q}_a\|_2 \|\mathbf{q}_b\|_2}$$

Contrastive loss



SimCLR's published architecture

Paper $h \rightarrow$ course ψ

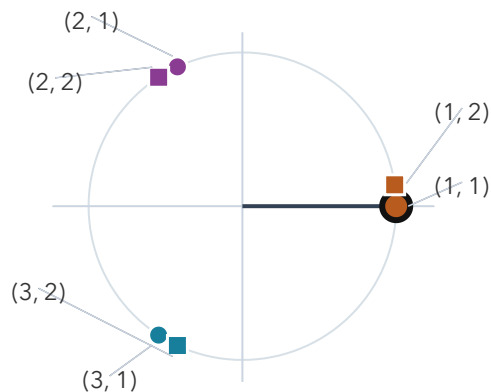
Paper $z \rightarrow$ course q

After pretraining: keep $f\theta$.
Evaluate its encoder features.

Chen et al. · ICML 2020 · Fig. 2

How the objective changes representations

Augmented view $\rightarrow \mathbf{f}\theta \rightarrow \psi \rightarrow \mathbf{g}\varphi \rightarrow \mathbf{q}$



Candidate (i, v)

(1, 2) · positive

(2, 1) · negative

(2, 2) · negative

(3, 1) · negative

(3, 2) · negative

cosine

0.990

-0.423

-0.545

-0.545

-0.423

Normalized probability



82.6%



4.9%



3.8%



3.8%

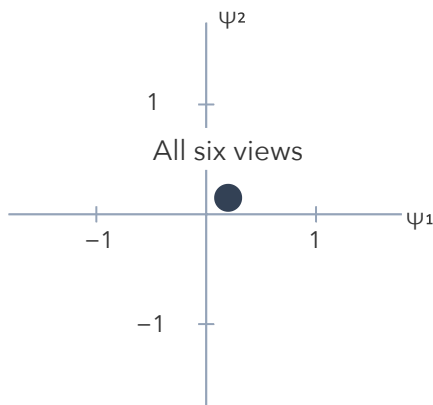


4.9%

$$\tau = 0.50 \cdot \text{positive probability} = 82.6\% \cdot \ell = 0.192 \cdot \text{mean over six anchors} = 0.189$$

Increase agreement with the positive relative to the competing views.

Constant representation: Collapse



Three original observations · two views each

Mean pair-agreement loss: **0.0000**

Feature spread: (0.000, 0.000)

One constant vector erases distinctions between observations.

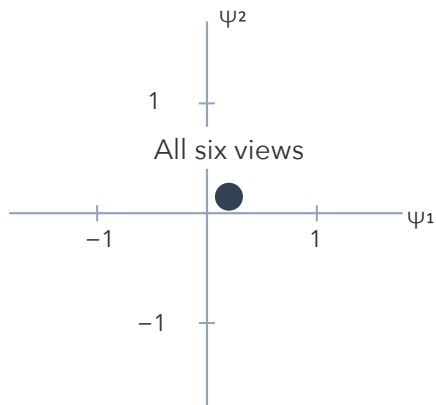
$\mathbf{c} = (0.20, 0.15)$

$$f_{\theta}(x) = \mathbf{c} \text{ for every } x \implies \ell_{\text{agree},i} = 0$$

The learning objective must avoid collapse of the representation as a possible optimal solution.

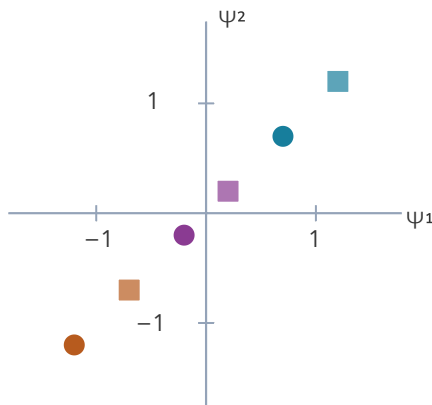
Loss of feature dimensions

Point: complete collapse



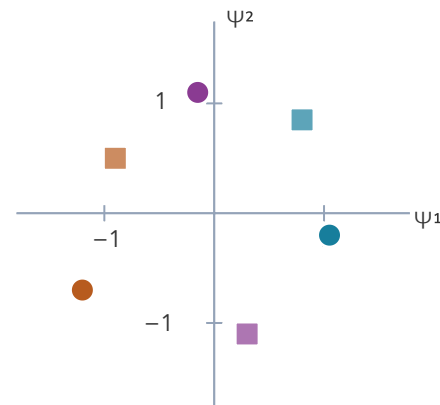
Std: (0.00, 0.00)

Line: one varying direction



Std: (0.89, 0.89)

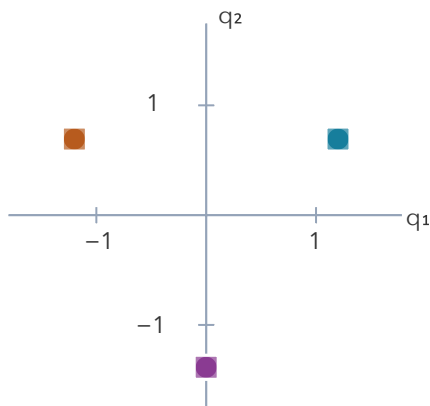
Two varying directions



Std: (0.91, 0.88)

Regularization: Variance-Invariance-Covariance (VICReg)

View $\rightarrow f\theta \rightarrow \psi \rightarrow \mathbf{g}\phi$ (expander) $\rightarrow \mathbf{q}$ · regularize $\mathbf{q}$ in both branches



Invariance · 0.0000

Distance between corresponding views

Variance · 0.0000

Per-dimension standard-deviation floor

Covariance · 0.0000

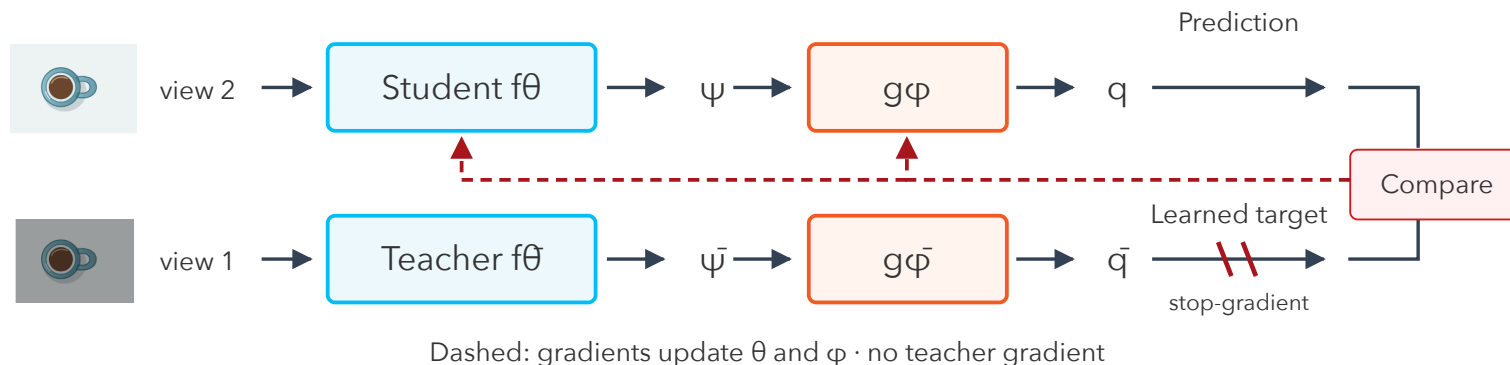
Squared off-diagonal covariance

$$\mathcal{L}_{\text{VICReg}} = \lambda_{\text{inv}} \mathcal{L}_{\text{inv}} + \lambda_{\text{var}} \mathcal{L}_{\text{var}} + \lambda_{\text{cov}} \mathcal{L}_{\text{cov}}$$

Covariance discourages redundant feature directions.

Bardes et al. · VICReg · ICLR 2022

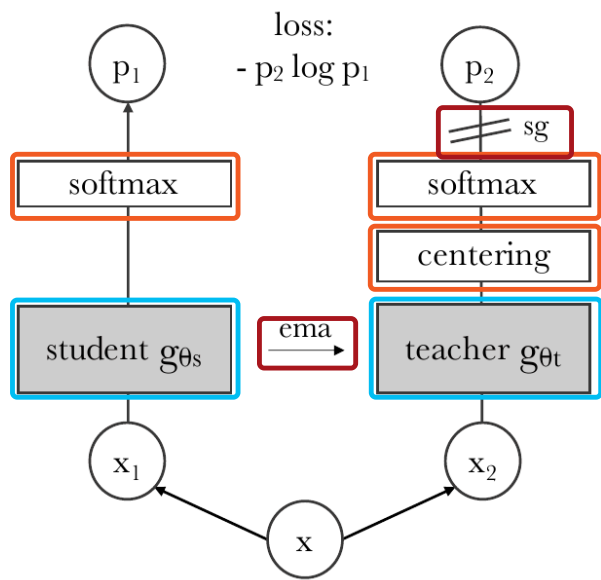
Self-distillation: Learning targets from a teacher



A target that evolves during learning

The student predicts a teacher output from a different view of the observation.

DINO architecture



Student and teacher include encoder + head

$$g_{\theta_s} \leftrightarrow g_{\phi} \circ f_{\theta}$$

$$g_{\theta_t} \leftrightarrow g_{\bar{\phi}} \circ f_{\bar{\theta}}$$

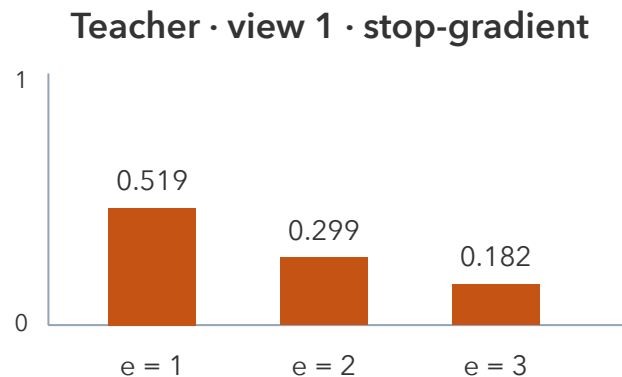
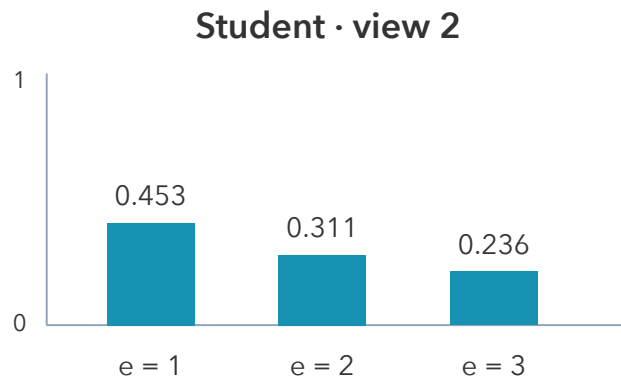
Center the teacher outputs, then apply separate softmax temperatures.

Stop the teacher gradient; update its parameters by a moving average.

Caron et al. · DINO · ICCV 2021 · main-paper Fig. 2

Match teacher and student distributions; update their parameters by different rules.

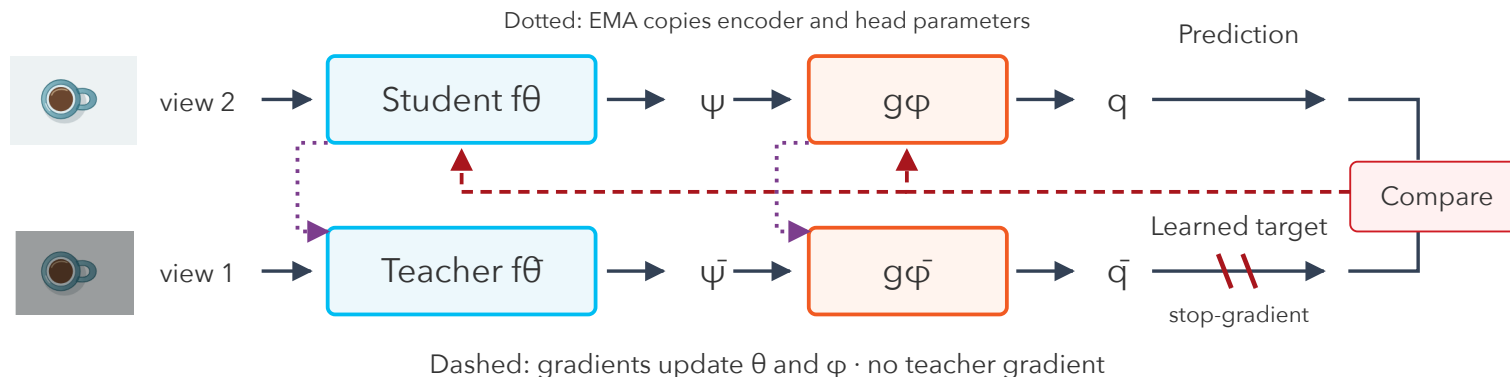
Distribution matching in DINO



$$\ell_{\text{DINO}} = - \sum_e \rho_{\text{teach},e} \log \rho_{\text{stu},e}$$

Computed one-view-pair cross-entropy: **1.0230**

Gradient updates and moving averages



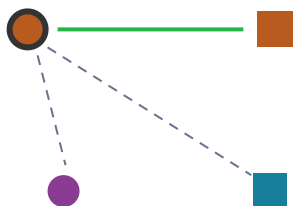
$$\bar{\theta}_{n+1} = \mu \bar{\theta}_n + (1 - \mu) \theta_{n+1}$$

Encoder: $\theta = 0.488 \cdot \bar{\theta} = 0.578$ Head: $\phi = 0.830 \cdot \bar{\phi} = 0.966$

After a batch: update the teacher head $\bar{\phi}$ using a moving average.

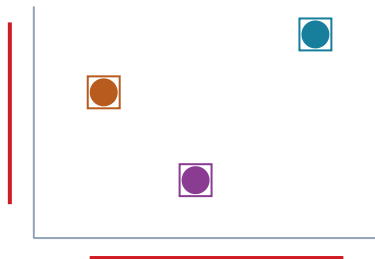
Summary of representation learning

SimCLR



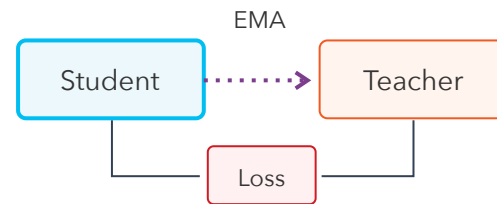
Competing views

VICReg



Spread + decorrelation

DINO



Centered, sharpened target

Teacher distribution

Observation $\rightarrow$ encoder $\rightarrow$ features $\rightarrow$ constructed relation or target $\rightarrow$ loss $\rightarrow$ parameter update

The goal of a good representation

Keep the information useful for the intended task.

Ignore the rest.

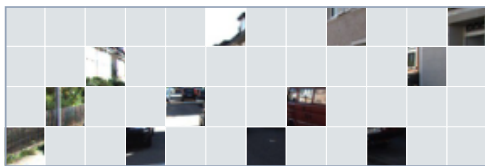
SECTION 04

Reconstruction and feature prediction



Reconstruction as a training signal

Partially observed input



Predicted pixels



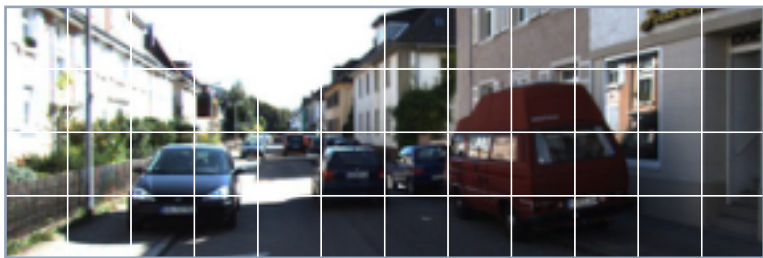
Original observed pixels supply the target.

Withhold information from the prediction input; keep it for the target.

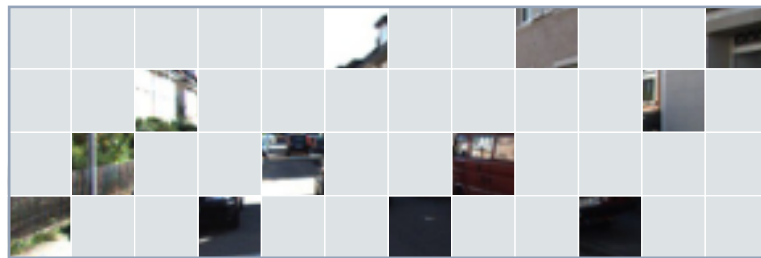
The original observation supplies a target without a human annotation.

Visible and masked patches

Original observation x



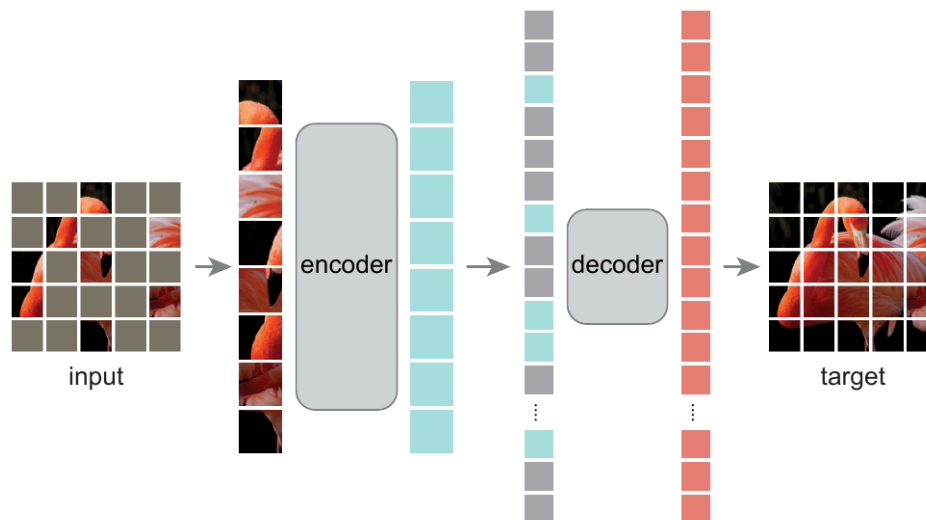
Visible patches V · hidden patches M



Visible patch tokens + their positions

The MAE encoder receives visible tokens, with their original positions.

MAE architecture

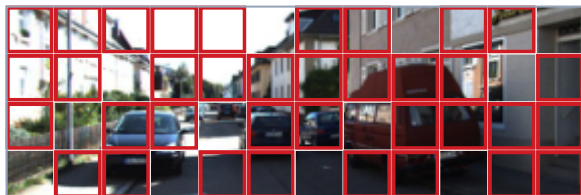


Mask tokens enter the decoder after visible-patch encoding.

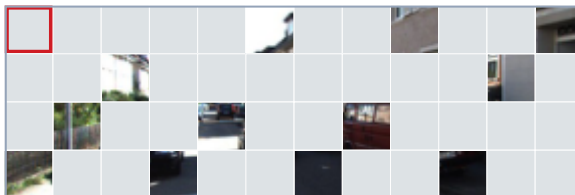
He et al., MAE, CVPR 2022 · Fig. 1 · arXiv v3

Reconstruction loss on masked locations

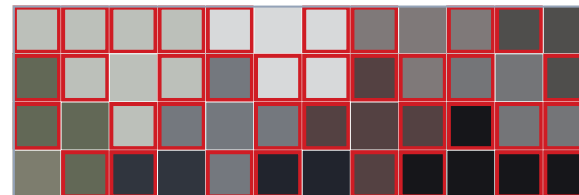
Original target pixels



Visible encoder input



Illustrative pixel prediction

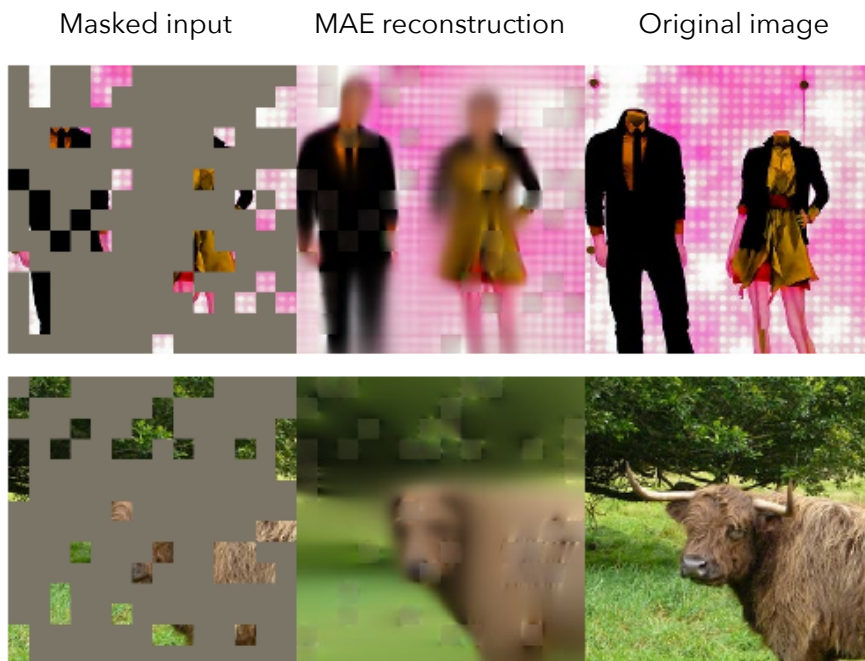


$$\mathcal{L}_{\text{pixel}} = \frac{1}{|\mathcal{M}|} \sum_{k \in \mathcal{M}} \|\hat{x}_k - x_k\|_2^2$$

Average over 36 masked patches: $\mathcal{L}_{\text{pixel}} = 148.32$

Visible patch errors are excluded; the basic loss acts on $\mathcal{M}$.

Plausible reconstruction and ambiguity



Multiple possible completions

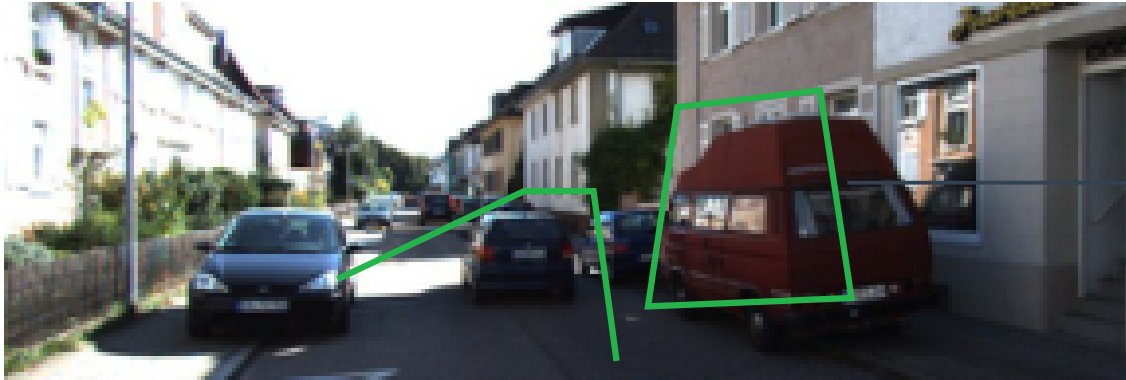
Visible context does not determine every hidden detail.

The outputs differ from the originals while retaining plausible scene structure.

Visual plausibility alone does not establish semantic understanding or robot-task accuracy.

He et al., MAE, CVPR 2022 · Fig. 3, two right-most examples · arXiv v3 · COCO validation images; model pretrained on ImageNet

What does a prediction target preserve?



Vehicle structure

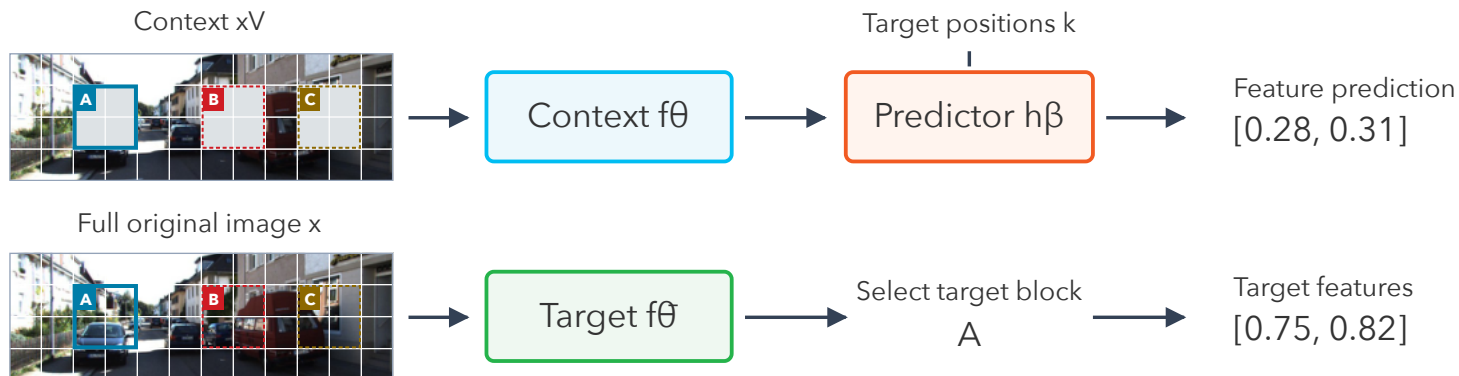
Potential task evidence

A robot task may place more value on object and road structure.

A target defines what prediction errors the model must reduce.

The features worth learning depend on the task.

JEPA: Predicting in feature space

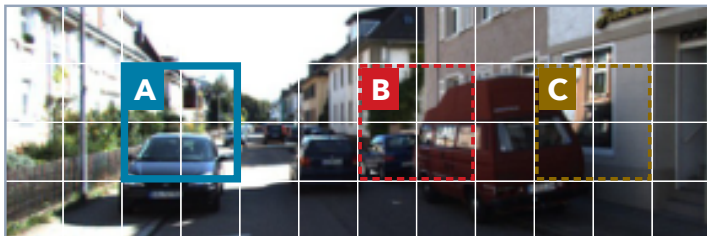


$$\psi_k^{\text{target}} = f_{\bar{\theta}}(x)_k$$

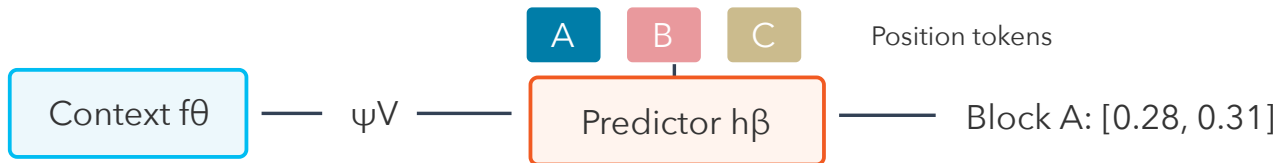
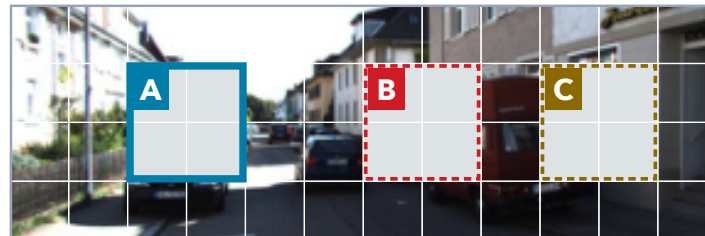
Compute the target in the intermediate feature space. No decoding into the output space (e.g., an image) is required. This supports learning more abstract concepts and more efficient training at scale.

JEPA: Context and target positions

Original · spatial target blocks A, B, C



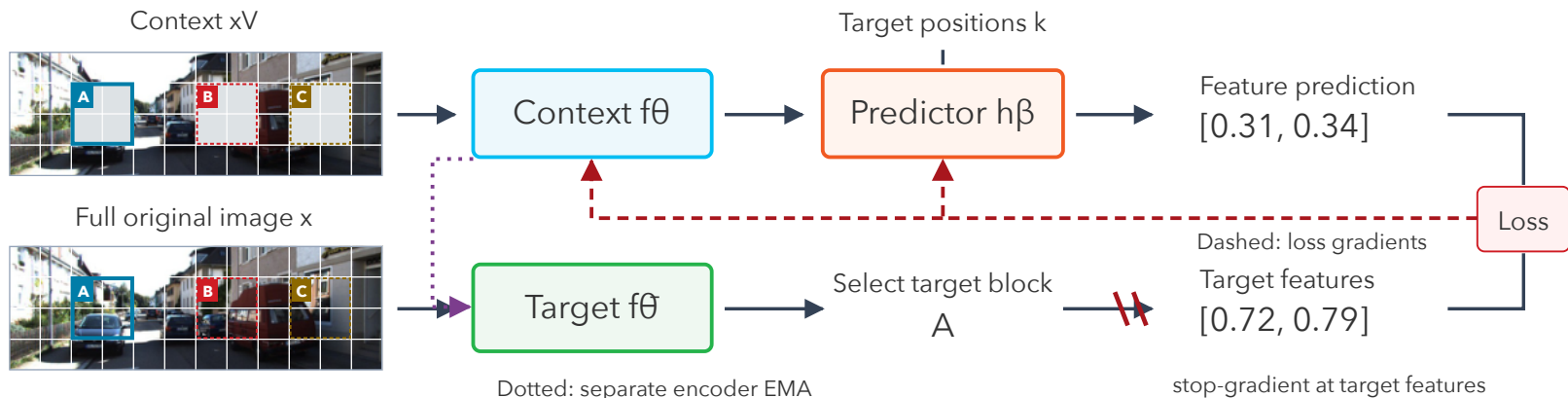
Context V · every target region removed



A position token identifies where to predict; it contains no hidden pixels.

$$\widehat{\psi}_k = h_\beta(f_\theta(x_V), k)$$

JEPA: Feature loss and target updates



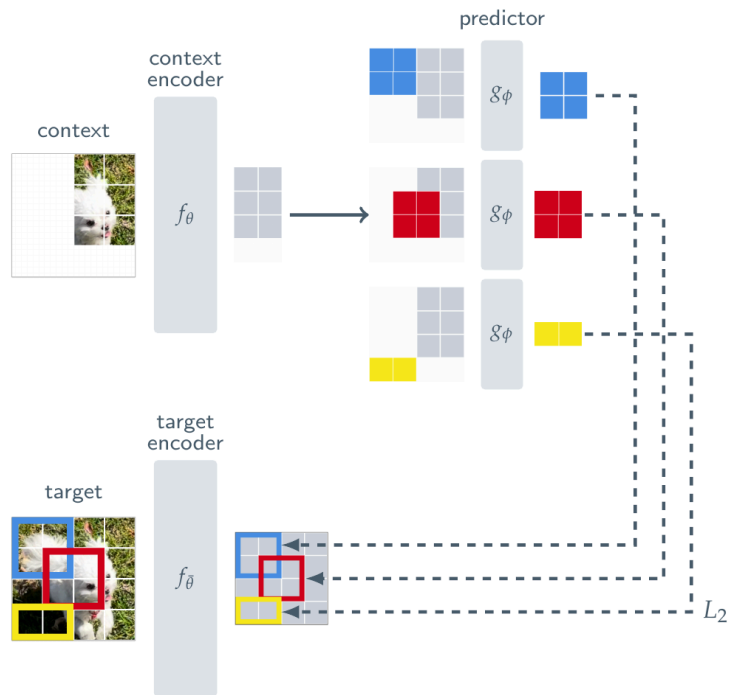
$$\bar{\theta}_{n+1} = \mu \bar{\theta}_n + (1 - \mu) \theta_{n+1}$$

L_{feature} = 0.1515

$\theta = 0.833 \cdot \beta = 0.738 \cdot \bar{\theta} = 0.967$

EMA transfers context-encoder parameters; the predictor has no EMA counterpart.

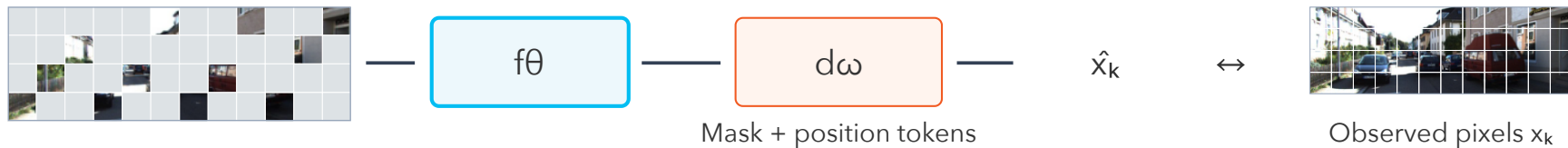
I-JEPA architecture



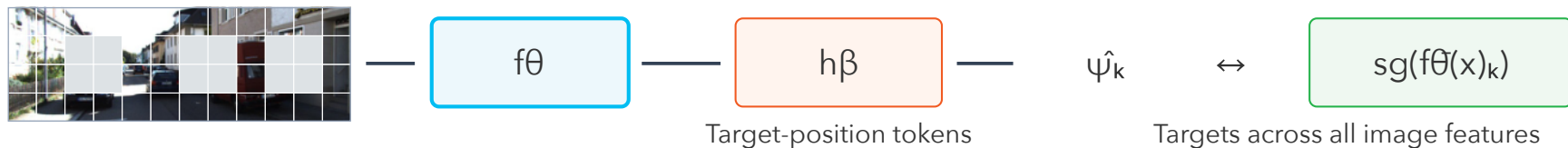
Assran et al., I-JEPA, CVPR 2023 · Fig. 3 · arXiv v3

Pixel targets and feature targets

MAE · pixel target



I-JEPA · feature target



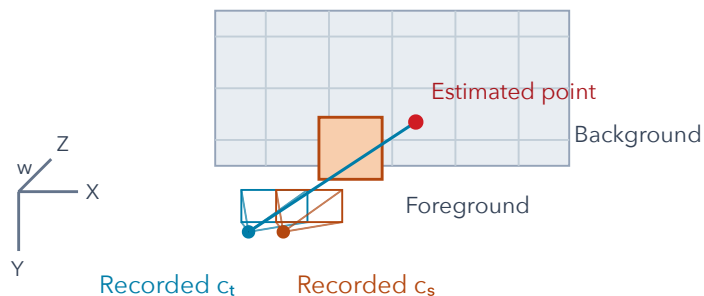
SECTION 05

Geometric supervision

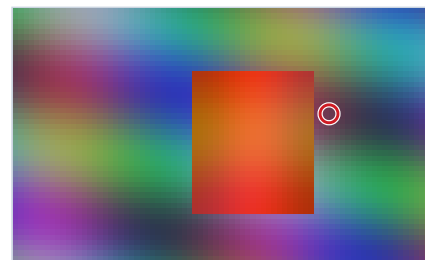


Learning depth from multiple observations

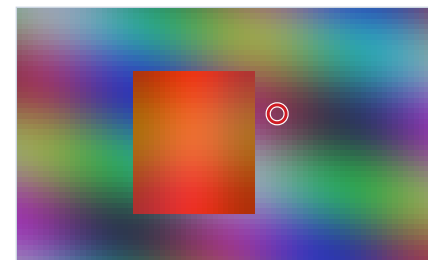
Synthetic 3D scene · fixed view



Recorded target $I_t \cdot u \rightarrow, v \downarrow$



Recorded source I_s



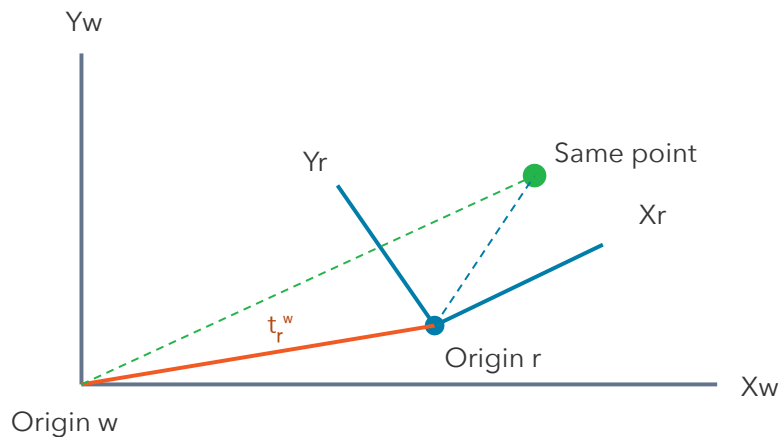
Source marker follows the computed projection.

$$D = 3.50 \cdot \hat{p} \text{ in } c_t: [1.06, -0.18, 3.50] \cdot u_s = [49.31, 20.00]$$

Recorded image values provide the target; depth and relative pose are predicted.

Scene structure and camera motion can turn recorded images into a learning signal.

The course's pose notation



Coordinates in robot frame

$p^r = [1.00, 0.60, 0.00]$

Coordinates in world frame

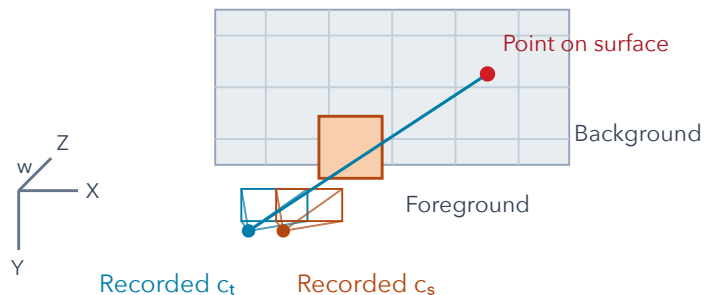
$p^w = [2.57, 1.42, 0.00]$

Translation: origin of r, expressed in w.

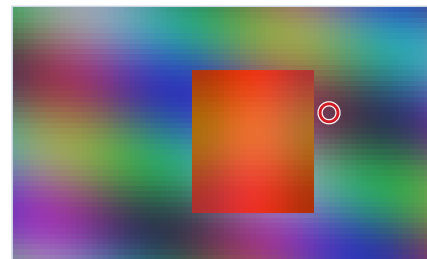
$$p^w = R_r^w p^r + t_r^w$$

Camera projection and intrinsics

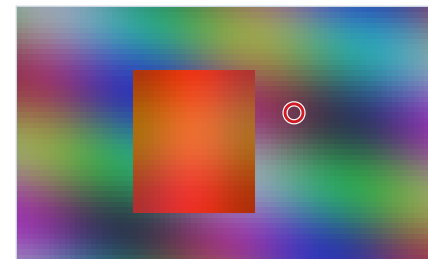
Synthetic 3D scene · fixed view



Recorded target $I_t \cdot u \rightarrow, v \downarrow$



Recorded source I_s



Source marker follows the computed projection.

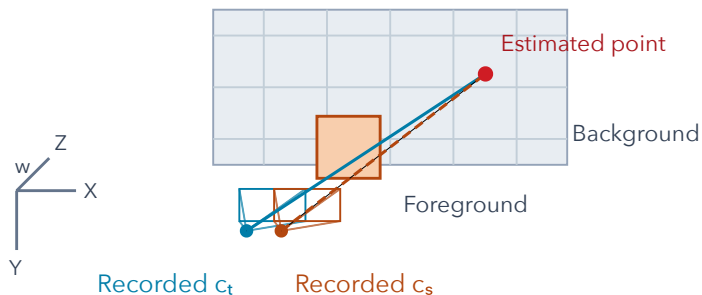
$$K_t = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

$f_x = f_y = 68 \cdot c_x = 39.5 \cdot c_y = 23.5 \cdot X$ right, Y down, Z forward

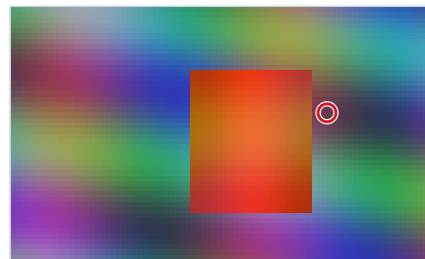
Optical-axis Z is depth; distance along a unit ray is a different quantity.

Back-projecting a target pixel

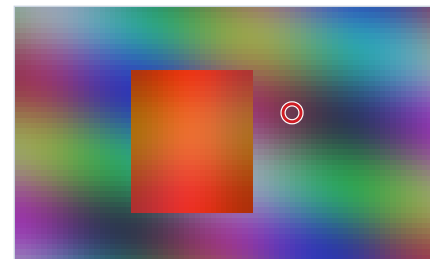
Synthetic 3D scene · fixed view



Recorded target $I_t \cdot u \rightarrow, v \downarrow$



Recorded source I_s



Source marker follows the computed projection.

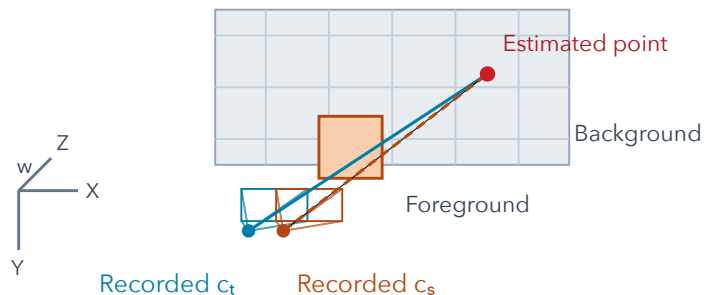
$$\hat{p}^{c_t} = \widehat{D}_t(\mathbf{u}) K_t^{-1} \tilde{\mathbf{u}}$$

$\tilde{D} = 5.00 \cdot \hat{p}$ in c_t : $[1.51, -0.26, 5.00] \cdot u_s = [52.52, 20.00]$

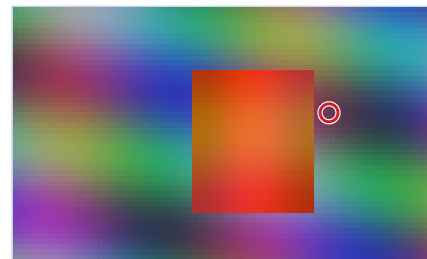
Changing estimated depth moves the hypothesis along a fixed target ray.

Changing camera frame

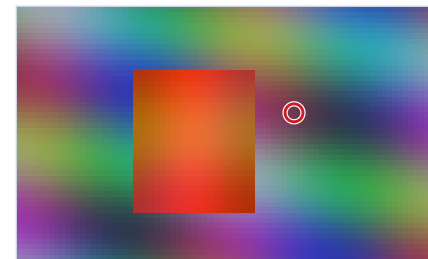
Synthetic 3D scene · fixed view



Recorded target $I_t \cdot u \rightarrow, v \downarrow$



Recorded source I_s



Source marker follows the computed projection.

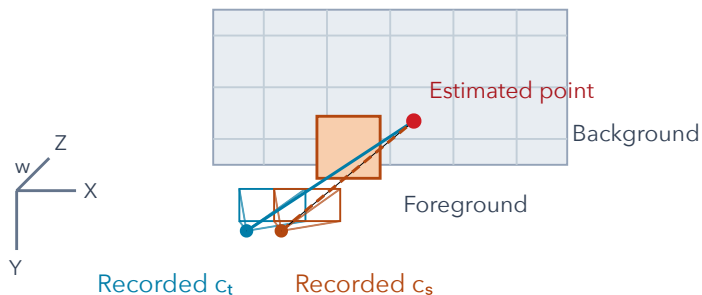
$$\hat{p}^{c_s} = \hat{R}_{c_t}^{c_s} \hat{p}^{c_t} + \hat{t}_{c_t}^{c_s}$$

$\hat{p}$ in c_t : [1.51, -0.26, 5.00] · $\hat{p}$ in c_s : [0.96, -0.26, 5.00]

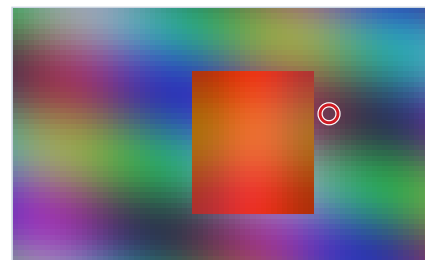
Changing the pose estimate changes coordinates and sampling; recorded images stay fixed.

Source projection and sampling

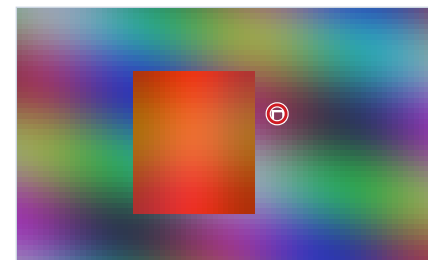
Synthetic 3D scene · fixed view



Recorded target $I_t \cdot u \rightarrow, v \downarrow$



Recorded source I_s



Target Sampled RGB L1 = 0.1420

Bilinear neighbors · weights

0.69 0.31 0.00 0.00

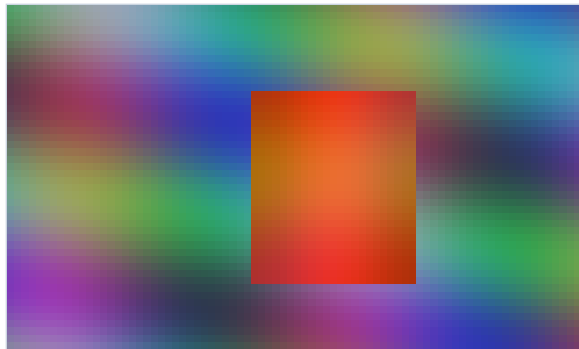
$$\hat{Z}_s > 0 \quad 0 \leq u_s \leq 79 \quad 0 \leq v_s \leq 47$$

$$D = 3.50 \cdot \hat{p} \text{ in } c_t: [1.06, -0.18, 3.50] \cdot u_s = [49.31, 20.00]$$

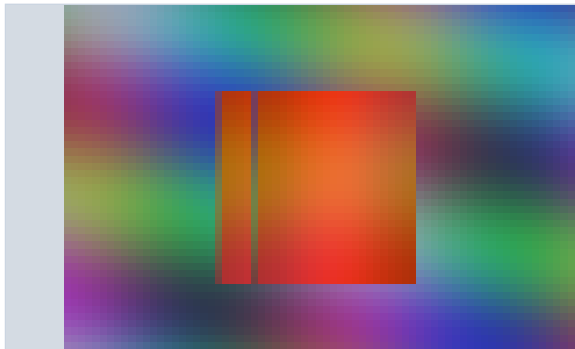
Bilinear interpolation combines neighboring recorded source values.

Photometric reconstruction loss

Recorded target I_t



Reconstructed target $\hat{I}_t$



RGB L1 error · brighter = larger

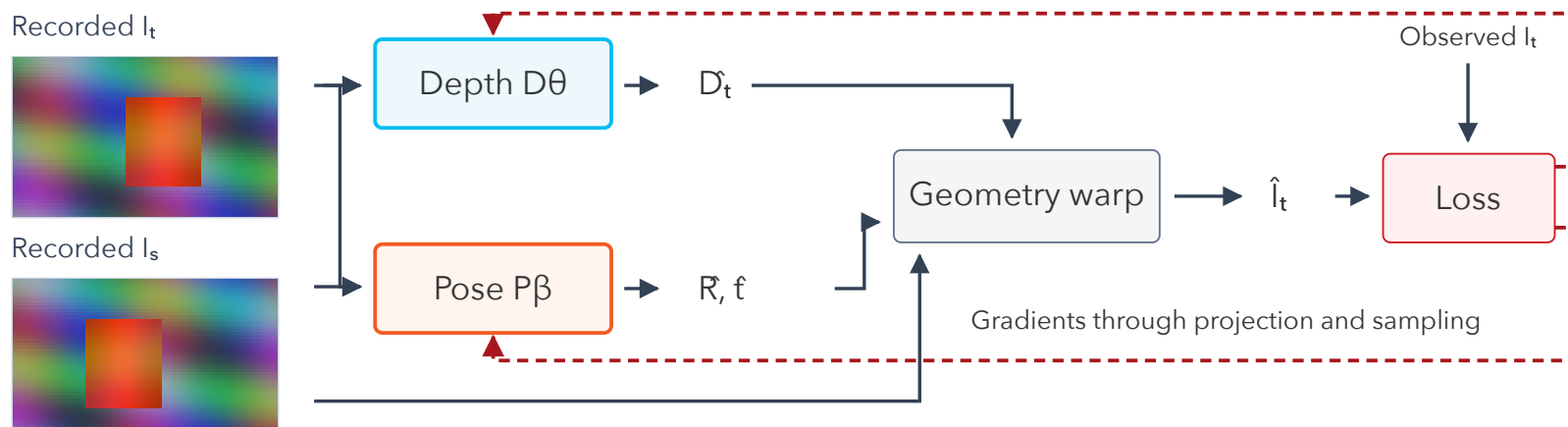


$$\mathcal{L}_{\text{photo}} = \frac{1}{|\Omega|} \sum_{\mathbf{u} \in \Omega} \|I_t(\mathbf{u}) - \hat{I}_t(\mathbf{u})\|_1$$

Mean RGB L1 = $0.0448 \cdot 3456 / 3840$ pixels in Ω

Ω excludes behind-camera and out-of-image projections; occlusions are still included.

Updating depth and pose networks



$$\Theta_{n+1} = \Theta_n - \eta \nabla_{\Theta} \mathcal{L}_{\text{photo}}, \quad \Theta = (\theta, \beta)$$

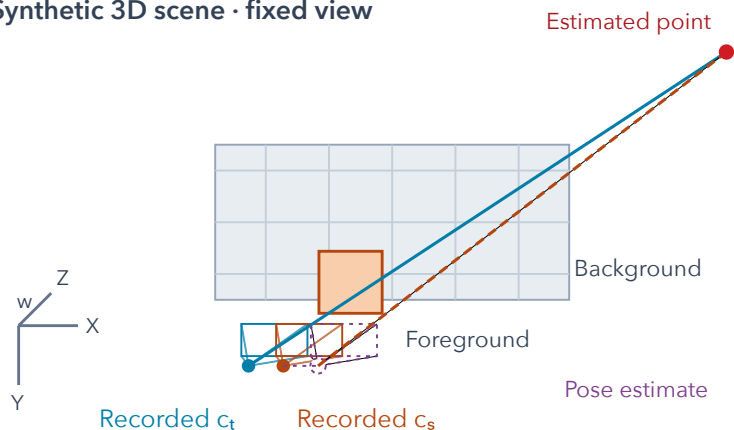
$\theta = \bar{D} = 3.504 \cdot \beta$ = baseline estimate = 0.524

One-pixel RGB L1 = 0.1181

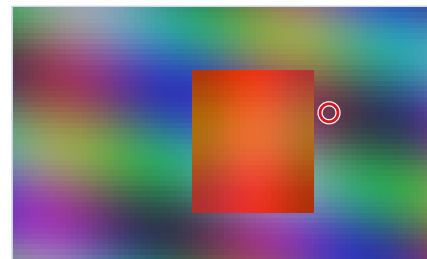
Computed scalar-model update: lower pixel error does not determine metric scale.

Monocular scale ambiguity

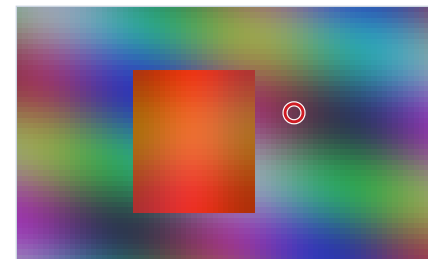
Synthetic 3D scene · fixed view



Recorded target $I_t \cdot u \rightarrow, v \downarrow$



Recorded source I_s



Target Sampled RGB L1 = 0.0004

Source marker follows the computed projection.

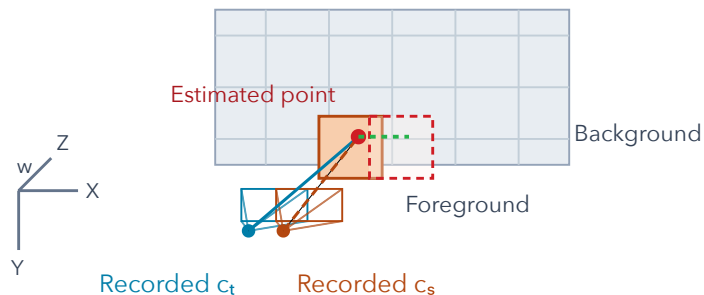
$$\pi(K_s \kappa \hat{p}^{c_s}) = \pi(K_s \hat{p}^{c_s}), \quad \kappa > 0$$

$D = 10.00 \cdot \hat{p}$ in c_t : $[3.01, -0.51, 10.00] \cdot u_s = [52.52, 20.00]$

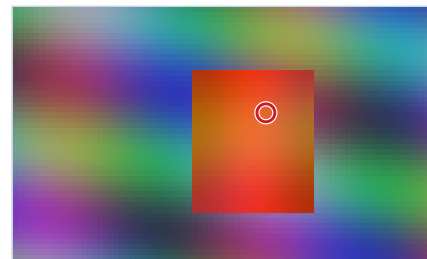
Scale the depth and every relative-translation component together; pixels stay fixed.

Independently moving objects

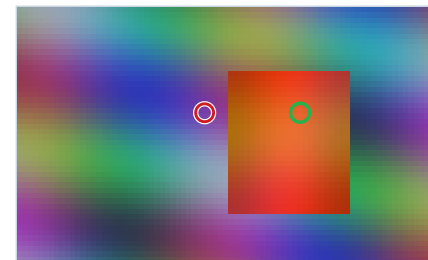
Synthetic 3D scene · fixed view



Recorded target $I_t \cdot u \rightarrow, v \downarrow$



Recorded source $I_s \cdot$ object moves



Target Sampled **RGB L1 = 1.0106**

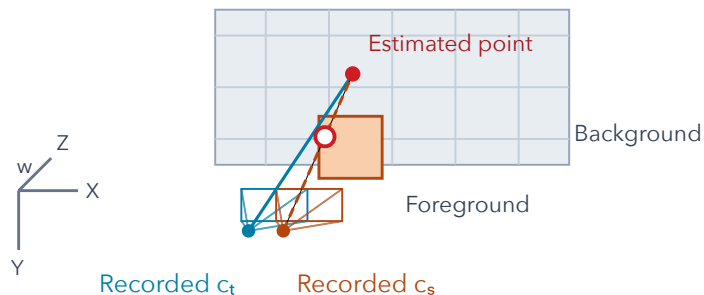
Red: rigid warp · green: actual moving-point projection

Rigid warp $u_s = [35.53, 20.00]$ · moving point $u_s = [53.67, 20.00]$

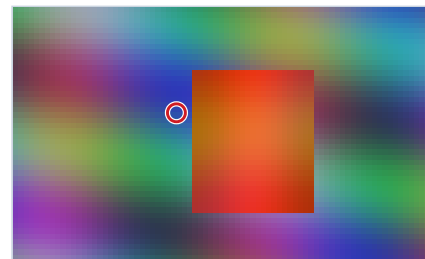
The moving object violates the static-scene assumption; its rigid warp is inconsistent.

Occlusion and disocclusion

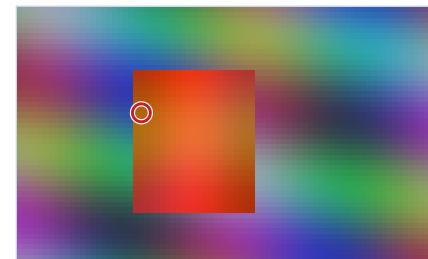
Synthetic 3D scene · fixed view



Recorded target $I_t \cdot u \rightarrow, v \downarrow$



Recorded source I_s



Target Sampled RGB L1 = 1.2332

Source marker follows the computed projection.

$Z_s = 5.00 > 0 \cdot u_s = [23.52, 20.00]$ · source ray hits the foreground first

A positive, in-bounds projection can still hit an occluded surface.

Texture and reflectance failures



Appearance consistency can fail

Top: distorted, reflective, and color-saturated regions.

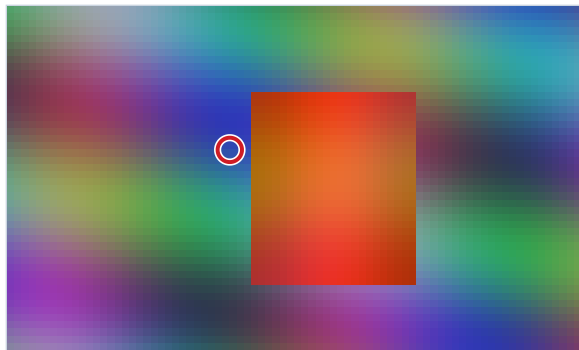
Bottom: ambiguous boundaries and intricate shapes can also be difficult.

These are the paper's examples, with its original method labels and highlighted regions.

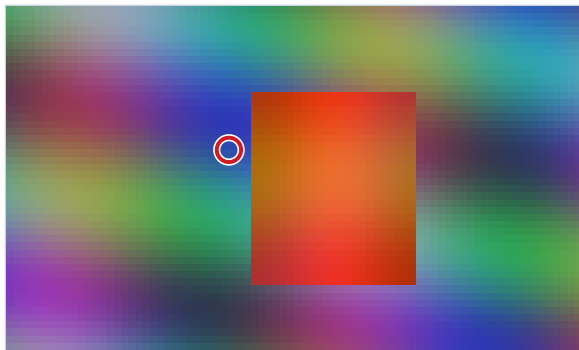
Godard et al., Monodepth2, ICCV 2019 · Fig. 8 · arXiv v4

Monodepth2's reconstruction choices

Recorded target I_t

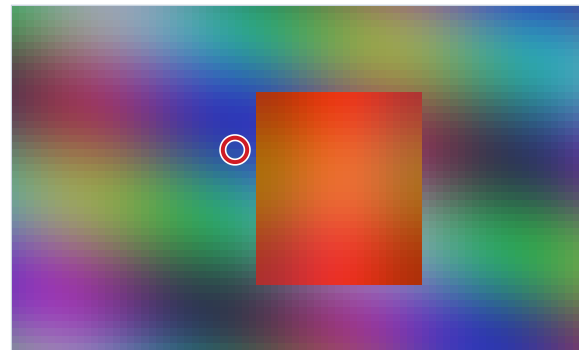


Recorded source A



Warped RGB L1: 0.0000

Recorded source B



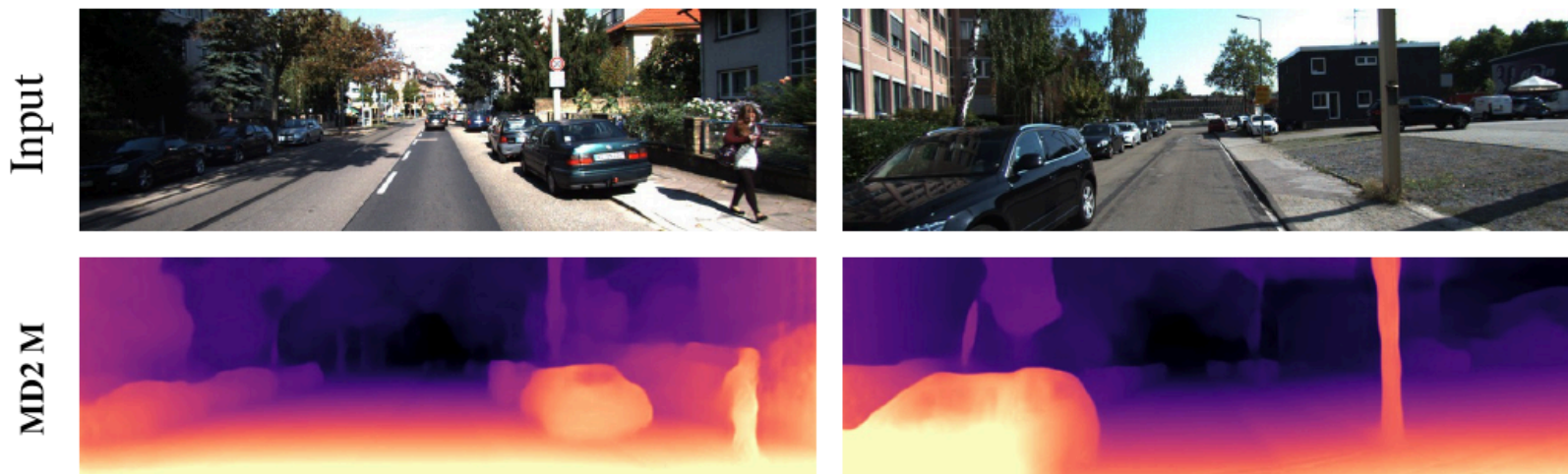
Warped RGB L1: 0.0000

$$m(\mathbf{u}) = \left[\min_s e_s^{\text{warp}}(\mathbf{u}) < \min_s e_s^{\text{identity}}(\mathbf{u}) \right]$$

Warp minimum 0.0000 · unwarp minimum 0.0000 · m = 0

Identical views provide no improvement from warping: the strict comparison rejects this example.

Learned depth on real images



Recorded camera images above; Monodepth2 monocular depth predictions below.

Training relates observations; inference predicts depth from one image.

Godard et al., Monodepth2, ICCV 2019 · Fig. 7, first two examples, Input and MD2 M rows · KITTI Eigen split · arXiv v4

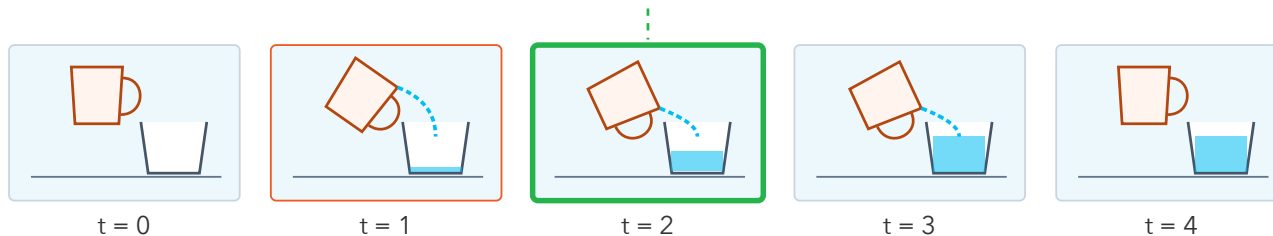
SECTION 06

Correspondence through space and time

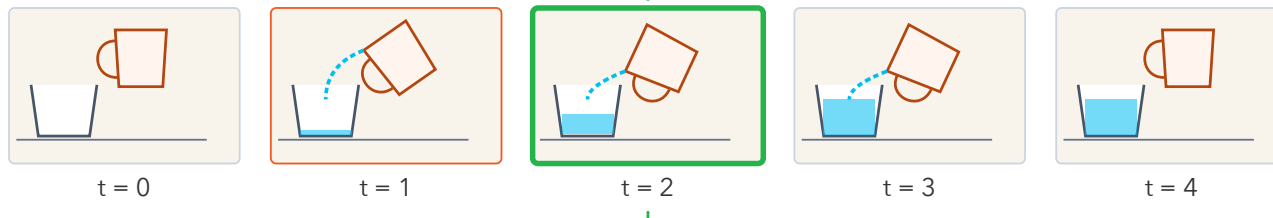


Time as a source of related observations

Camera A

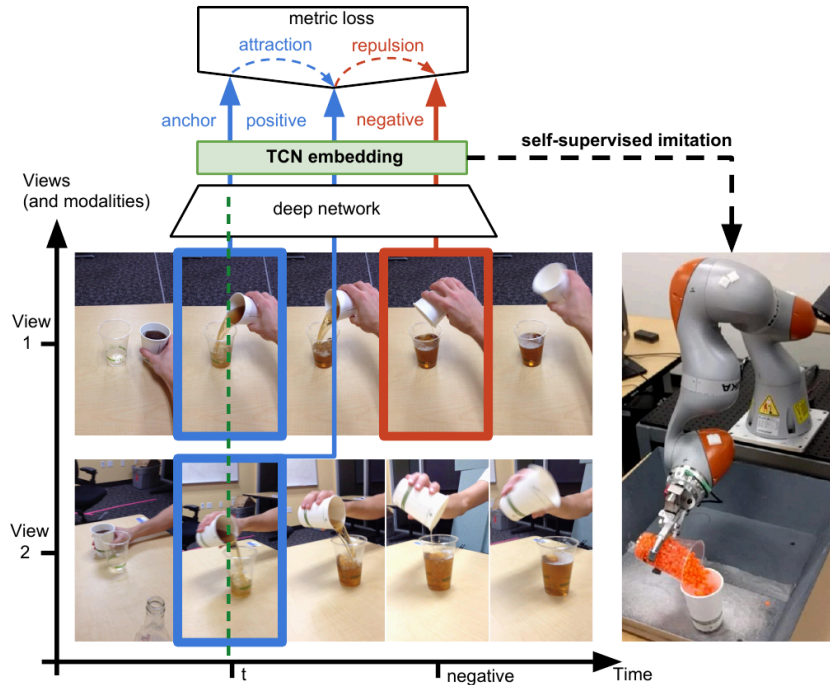


Camera B



Adjacent timestamps can have different tilt, flow, and recipient fill.

Time-Contrastive Networks



Sermanet et al., Time-Contrastive Networks, ICRA 2018 · Fig. 1 · arXiv v3

Anchor and positive

Different views of the same timestamp.

Temporal negative

A different time in the same sequence.

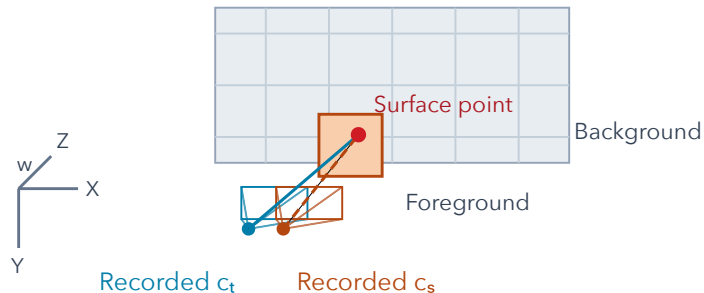
Green cursor links the simultaneous views. Original blue and red boxes retain the paper's roles.

Video relations supervise this objective.

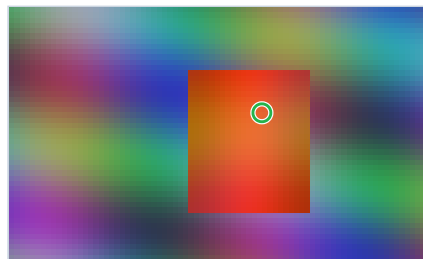
The experimental encoder starts from ImageNet weights.

Geometrically generated pixel matches

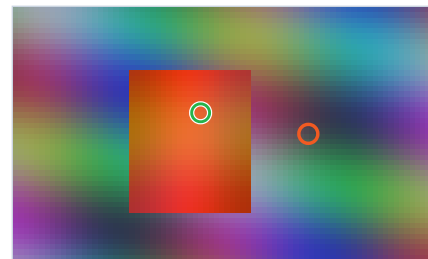
Synthetic 3D scene · fixed view



Target I_t



Source I_s



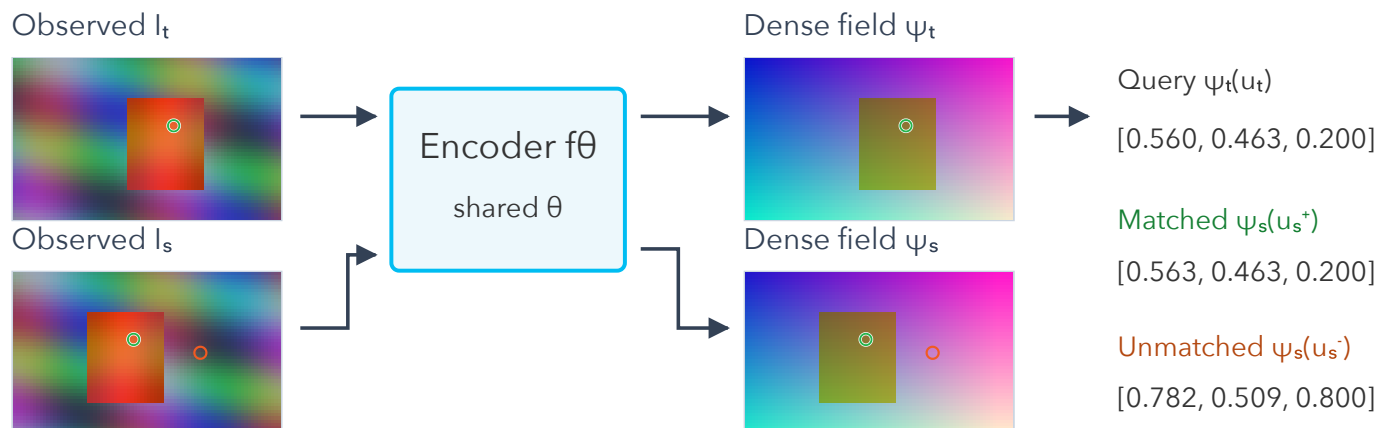
Green: same surface point · orange: different surface point

$$p^{c_t} = D_t(\mathbf{u}_t) K_t^{-1} \tilde{\mathbf{u}}_t, \quad \mathbf{u}_s = \pi(K_s(R_{c_t}^{c_s} p^{c_t} + t_{c_t}^{c_s}))$$

$\mathbf{u}_t = [48.00, 20.00]$ · projected $\mathbf{u}_s = [35.53, 20.00]$ · visible match available

A nonmatch is a different visible point, not merely another camera view.

Learning dense descriptors



$$d_+ = \|\psi_t(\mathbf{u}_t) - \psi_s(\mathbf{u}_s^+)\|_2, \quad d_- = \|\psi_t(\mathbf{u}_t) - \psi_s(\mathbf{u}_s^-)\|_2$$

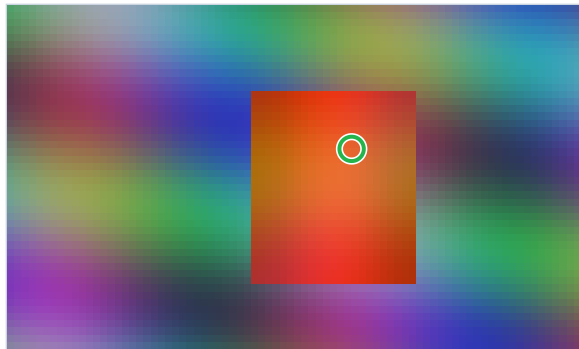
Matched distance $d_+ = 0.0033$

Unmatched distance $d_- = 0.6414$

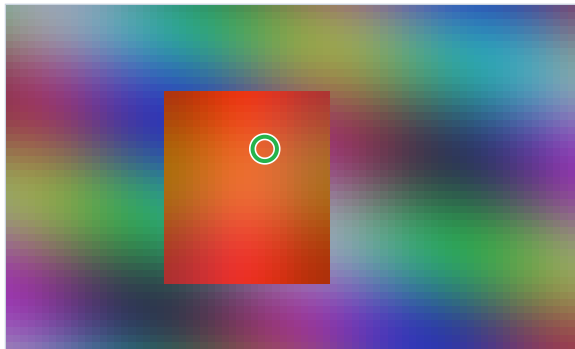
Nonmatching pairs provide a separation constraint; geometry supplies the pair roles.

Matching with learned descriptors

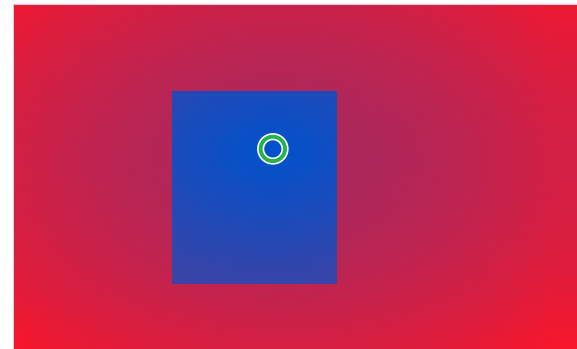
Query image I_t



Search image I_s



Distance · blue is closer



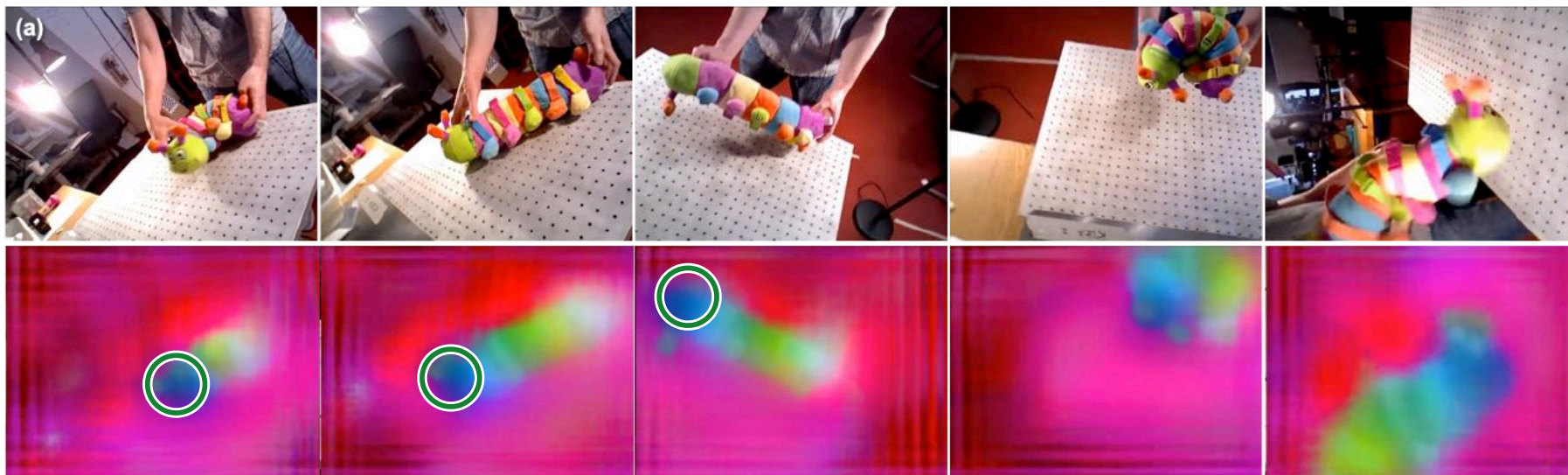
$$\mathbf{u}_s^* = \arg \min_{\mathbf{u}_s} \|\psi_t(\mathbf{u}_t) - \psi_s(\mathbf{u}_s)\|_2$$

Compared 3840 / 3840 descriptors · best candidate [36.00, 20.00] · distance 0.0033

Nearest-feature search returns a candidate, not a guarantee of visibility or grasp validity.

Descriptor consistency across configurations

RGB observations above · dense descriptor colors below

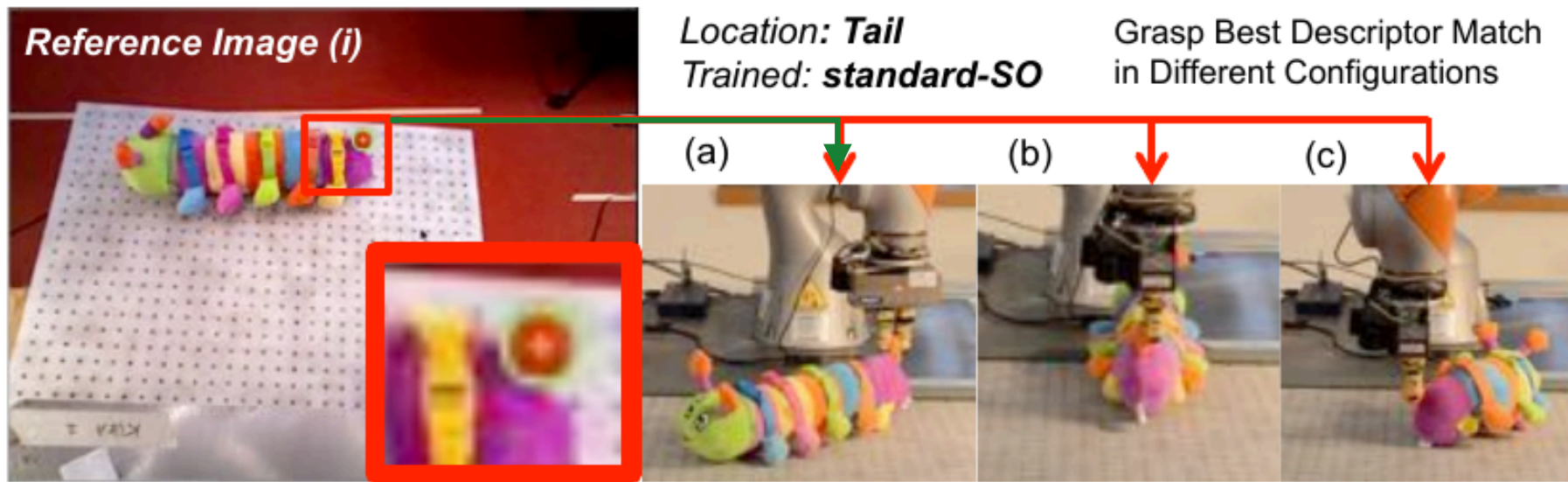


The same object changes configuration while corresponding regions retain similar feature colors.

Feature colors encode descriptors, not manually assigned semantic part labels.

Florence et al., Dense Object Nets, CoRL 2018 · Fig. 2(a), all five configurations · arXiv v2

Transferring a grasp point



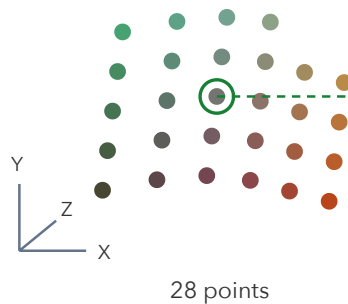
User specifies a task pixel → Descriptor match in a new image → Depth + grasp geometry

The matched image point is a perception input to grasp planning.

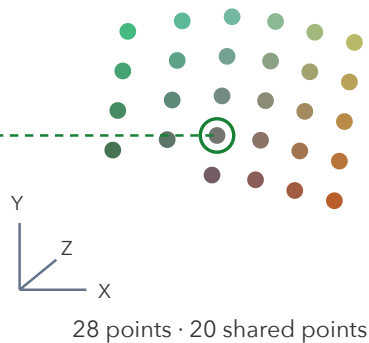
Florence et al., Dense Object Nets, CoRL 2018 · Fig. 6(i), tail, standard-so · arXiv v2

Extension to 3D correspondence

Cloud A · frame a



Cloud B expressed in a



$$p^a = R_b^a p^b + t_b^a$$

Query $p^a = [-0.18, -0.15, 0.49]$ · matched $p^b = [0.54, -0.48, 0.79]$

Several reliable, well-distributed correspondences can constrain a rigid registration.

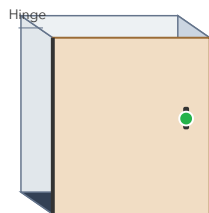
SECTION 07

Learning action possibilities from experience



Observation, action, outcome, target

Earlier observation x_t



Opening: 0.0°

$t = 0$ · before the action

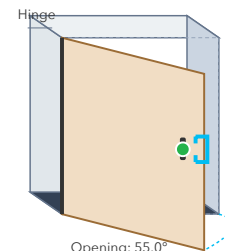
Commanded action a_t



Grasp handle, pull outward

Desired opening: $\geq 30^\circ$

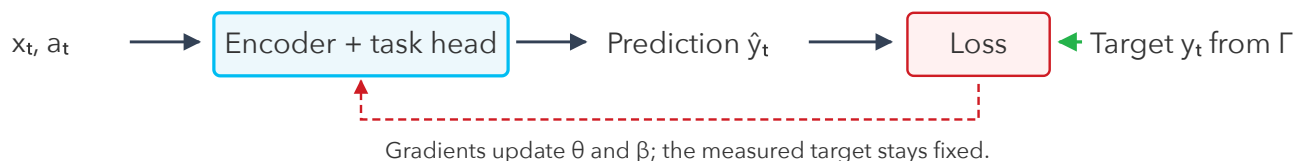
Later sensor measurements o



Opening: 55.0°

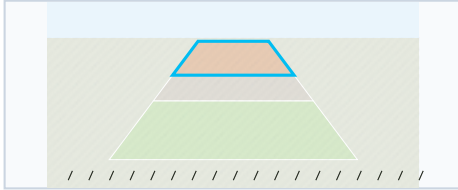
Contact detected · opening 55°

$$y_t = \Gamma(x_t, a_t, o_{t:t+\Delta})$$



Illustrative prediction 0.881 · measured target 1 · binary loss 0.1269

Traversability depends on the robot



Same selected mud patch · same 1.00 m/s command



Narrow wheels / controller A

Measured 0.25 m/s

Automatic cost: 0.75



Broad tracks / controller B

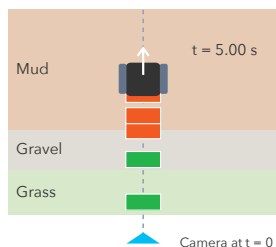
Measured 0.65 m/s

Automatic cost: 0.35

Traversability depends on the platform, controller, action, and outcome rule.

Measuring traversal outcomes

Visited ground · top view



Earlier camera image $l_0 \cdot t = 0$

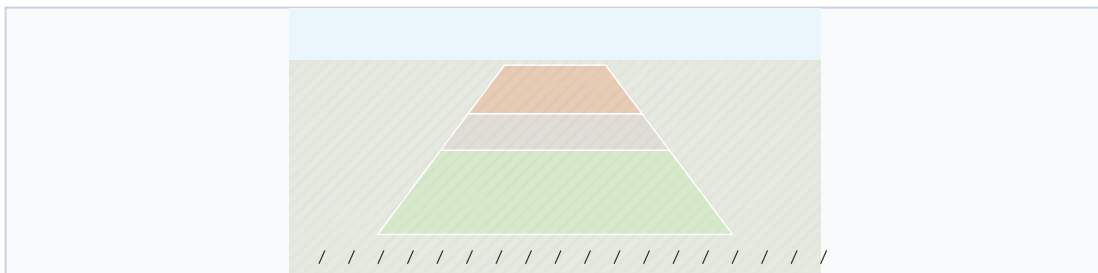


Image recorded → Outcome now available

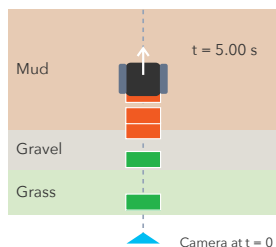
$$y = \Gamma(o) = \text{clip}_{[0,1]}(1 - v_{\text{meas}}/v_{\text{cmd}})$$

Commanded 1.00 m/s · measured 0.60 m/s · automatic cost $y = 0.40$

This run measures motion; the chosen proxy converts it into a cost.

Delayed supervision

Visited ground · top view



Earlier camera image $l_0 \cdot t = 0$

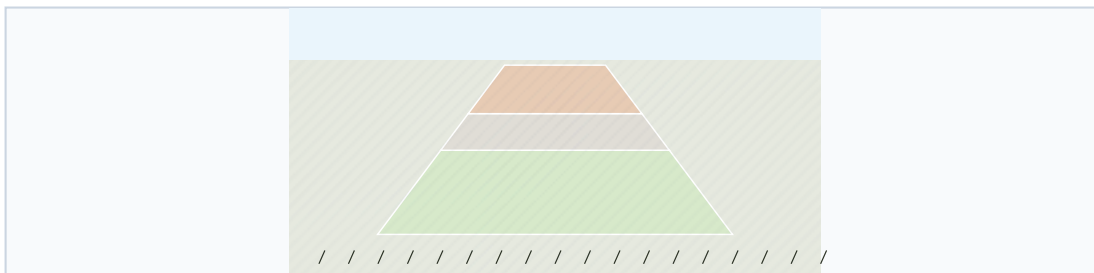


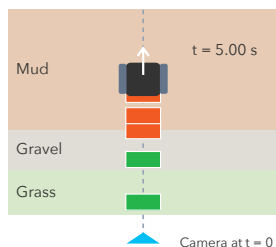
Image recorded → Outcome now available

Commanded 1.00 m/s · measured 0.60 m/s · automatic cost $y = 0.40$

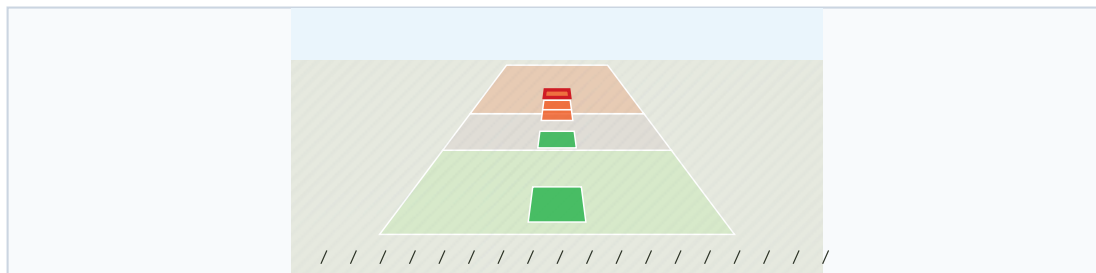
The earlier image is fixed. A target becomes available only after physical traversal.

Projecting experience into an earlier image

Visited ground · top view



Earlier camera image $l_0 \cdot t = 0$



■ Observed lower cost ■ Observed higher cost ▨ Unvisited

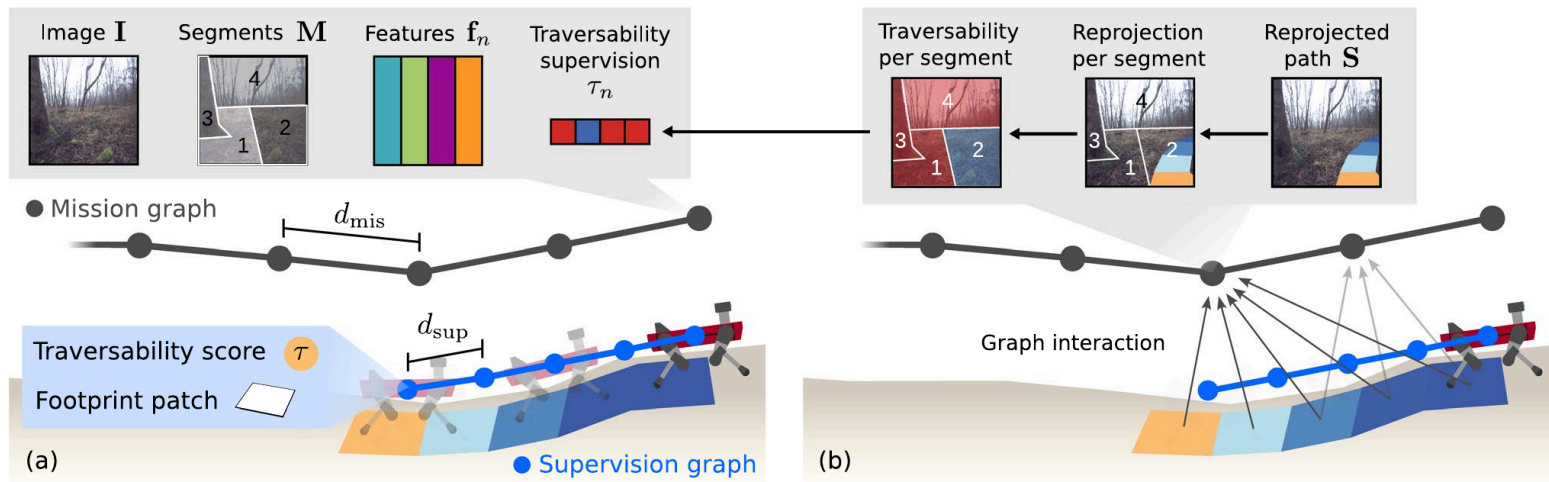
$$\mathbf{u}_0 = \pi(K_0(R_r^{c_0} p^r + t_r^{c_0}))$$

Commanded 1.00 m/s · measured 0.60 m/s · automatic cost $y = 0.40 \cdot 5$ projected targets · pose time offset 0.0 s

Toy local estimate: 0.500 → measured target 0.40 · mean squared error 0.0690

Restore time alignment before using these measured targets.

Wild Visual Navigation



Visual input

RGB images and pretrained visual features

Measured signal

Commanded versus estimated planar velocity

Accumulated supervision

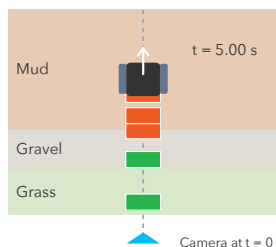
Footprint scores reprojected into stored images

Experienced motion supplies targets for visual traversability estimation.

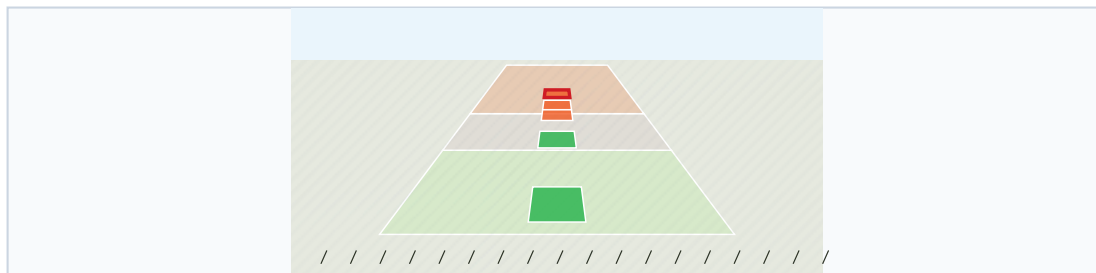
Mattamala et al., WVN, Autonomous Robots 2025 · Fig. 5(a,b), journal PDF page 6

Measured outcomes and unvisited terrain

Visited ground · top view



Earlier camera image $I_0 \cdot t = 0$



■ Observed lower cost ■ Observed higher cost ■ Unvisited

Commanded 1.00 m/s · measured 0.60 m/s · automatic cost $y = 0.40$

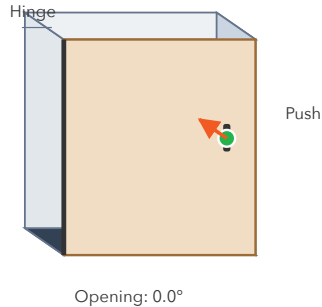
The driven path selects the training data; unvisited regions still have no measurement.

WVN's zero score for unvisited segments is a conservative modeling choice, not an observed failure.

Mattamala et al., WVN, 2025 · Sections 3.4.3 and 3.5.2

Manipulation affordances

Fixed initial observation x_t



Pull outward

Initial opening direction

Along the closed door

Little opening moment

Push inward

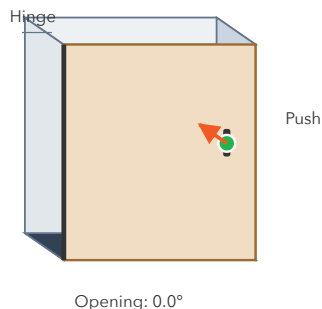
Closed stop blocks motion

Contact radius 0.85 m · direction $F = [0.00, 1.00, 0.00]$ · fixed grasp orientation

The same handle supports different outcomes for different action directions.

Predicting an action's outcome

Fixed initial observation x_t



Pull outward

Initial opening direction

Along the closed door

Little opening moment

Push inward

Closed stop blocks motion

Contact radius 0.85 m · direction $F = [0.00, 1.00, 0.00]$ · fixed grasp orientation

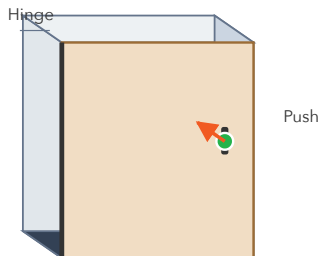
$$\hat{y}_t = h_{\beta}(f_{\theta}(x_t), a_t)$$

Predicted opening success: 0.018

Predict from the observation and proposed action before seeing the outcome.

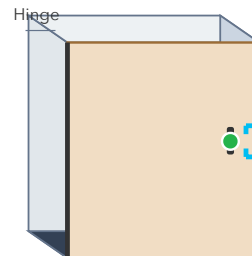
Automatically constructing an affordance target

Fixed initial observation x_t



Opening: 0.0°

Executed observation · complete



Opening: 0.0°

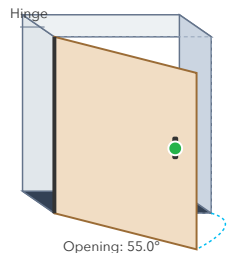
Contact radius 0.85 m · direction $F = [0.00, 1.00, 0.00]$ · fixed grasp orientation

$$y_t = \Gamma(o) = \mathbf{1}[\text{measured opening} \geq 30^\circ]$$

Predicted opening success: 0.018 · Measured opening $0.0^\circ \rightarrow$ automatic target $y = 0$

Task objective: open the door by at least 30° ; small movement alone is not success.

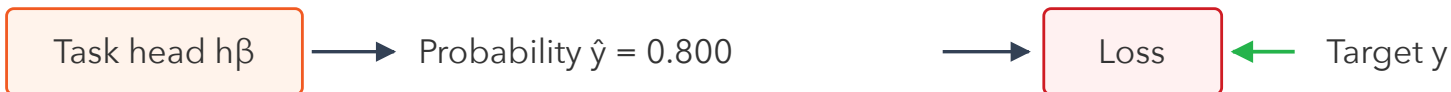
Losses for interaction-derived targets



Binary task outcome

Rule Γ : measured opening $\geq 30^\circ$

Automatic target $y = 1$



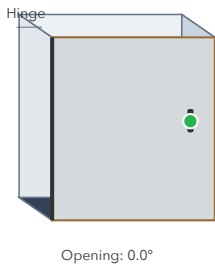
$$\hat{y} = \text{sigmoid}(b), \quad \ell = -y \log \hat{y} - (1 - y) \log (1 - \hat{y})$$

Computed binary cross-entropy: 0.2231

Choose a loss for the target type; self-supervision comes from how y was obtained.

ActAIM as an interaction-learning example

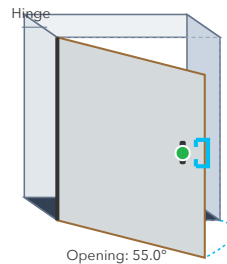
Initial depth D_0



Execute action in simulation

- Contact point p
- Gripper orientation R
- Motion direction F

Later depth D_1



Encode before / after depth → Embedding-change threshold → **Automatic interaction label**

Learn interaction modes

Latent model + local geometry

→

Propose actions

Where and how to interact

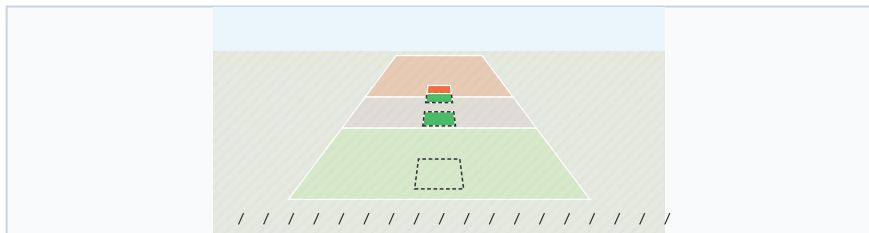
The paper learns a distribution of interactions that produce visual change.

A change criterion does not by itself specify "open this door by 30°".

Wang et al., ActAIM, ICRA 2023 · Sections III, IV-A/B, V-A · arXiv v1

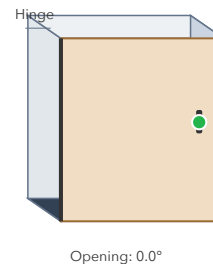
Imperfect targets and selected experience

Traversal: association error



Timestamp error $\rightarrow$ wrong image patch
 $\rightarrow$ a target for the wrong observation

Cabinet: measurement error



Actual opening 0° · sensor reports 35°
 $\rightarrow$ target $y = 1$ for an unopened door

A target can be generated automatically and still be wrong.

Attempted action records

Handle / pull: $y = 1$

Handle / push: $y = 0$

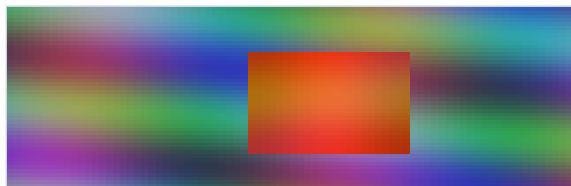
Other contact / direction: unattempted

The robot's selected experience leaves gaps in the training data.

The shared learning principle

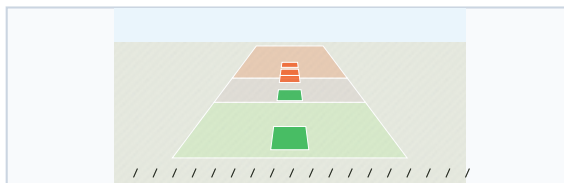
Geometric association

Recorded camera view



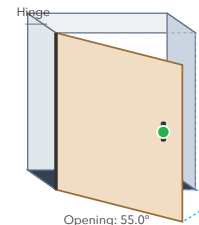
Depth + poses + visibility
→ corresponding pixels

Traversal experience



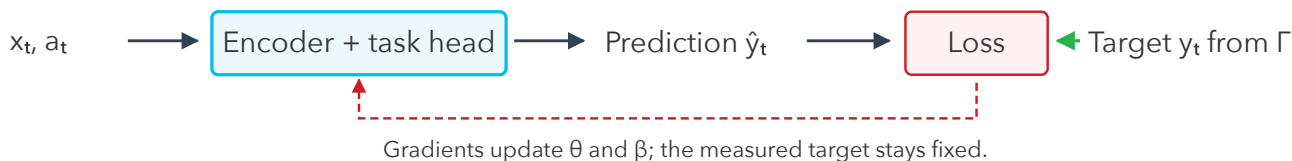
Command + measured motion
→ local cost targets

Cabinet interaction



Measured task outcome
→ success target

$$\hat{y}_t = h_{\beta}(f_{\theta}(x_t), a_t), \quad y_t = \Gamma(x_t, a_t, o_{t:t+\Delta})$$



Identify what supplies the target, which observations it applies to, and what the model must predict.

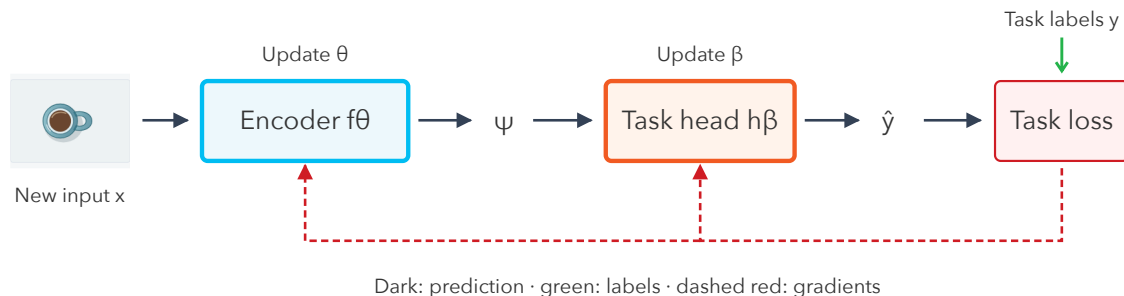
SECTION 08

Introductory adaptation and evaluation



Frozen encoder and fine-tuning

Fine-tune encoder + task head



$$\Theta = (\theta, \beta), \quad \text{update encoder and head}$$

Frozen features

Evaluate information accessible to the chosen head.

Fine-tuning

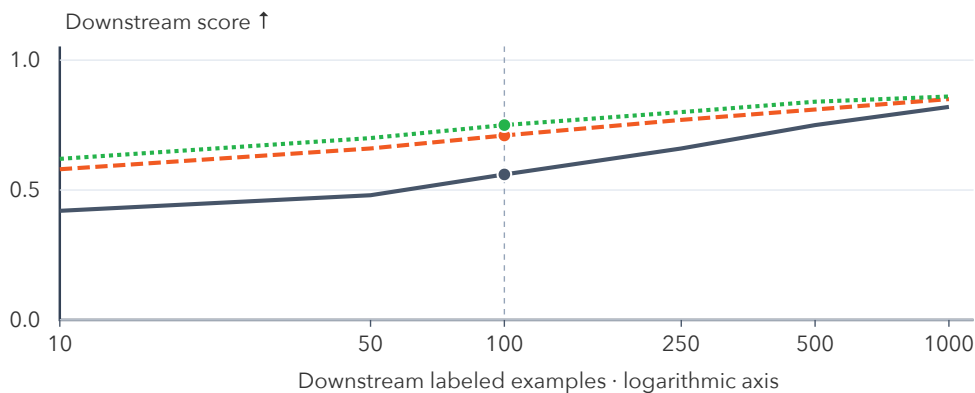
Adapt features and head to the target task.

For a linear probe, $h\beta$ is linear. A frozen encoder can also feed a nonlinear head.

Chen et al., SimCLR, ICML 2020 · Appendix B.8

Baselines and labeled-data efficiency

Illustrative curves · no measured result or universal ranking



— Random initialization **0.56**

- - Supervised pretraining **0.71**

... Self-supervised pretraining **0.75**

Same downstream split and task head · consistent tuning rules
Report pretraining data, annotations, and compute separately.

Compare label requirements at a stated task score under a declared evaluation protocol.

Splitting robot data correctly

T · Train

V · Validation

E · Test / evaluation

Time advances →



Train/test audit · adjacent pairs: 0 · shared trajectories: 0 · shared environments: 0

Whole environments are separated in these metadata; verify the actual recordings too.

Choose the split for the claim: new run, recording session, object, or environment.

[scikit-learn · grouped-data evaluation; robot grouping shown here is illustrative](#)

Evaluating the intended perception task

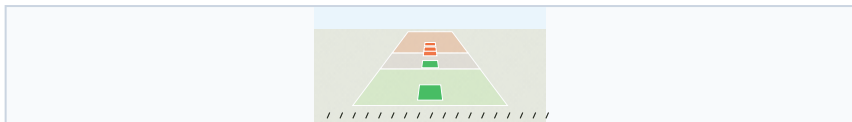
Depth

$$\text{AbsRel} = \text{mean}(|\hat{D} - D| / D)$$

Reference D	2 m	4 m	6 m
Prediction $\hat{D}$	2.0 m	4.0 m	6.0 m

AbsRel: **0.000** · median-aligned

Traversability



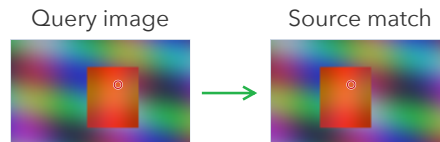
Predicted cost error on held-out experienced patches

Scale alignment uses reference depth at evaluation; it does not recover metric scale at inference.

Predictive error and closed-loop robot success answer different questions.

Monodepth2 official evaluator · AbsRel and monocular median scaling

Correspondence



Pixel error / match recall on visible held-out pairs

Manipulation affordance



Pull at handle

Predicted opening

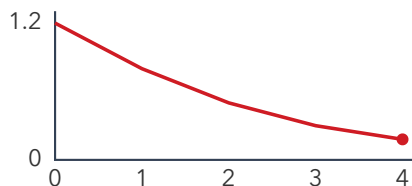
success: **0.881**

Held-out outcome classification or probability loss

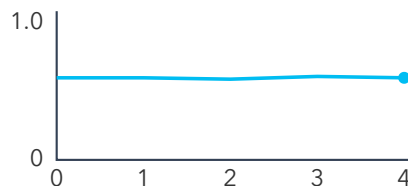
Pretext loss and downstream usefulness

Constructed example · these curves are not an experimental result

Training pretext loss ↓

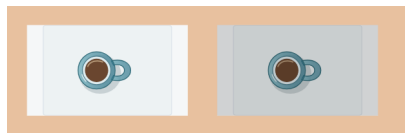


Held-out downstream score ↑



Horizontal axes: optimization checkpoint n

Optimization checkpoint $n = 4$ · pretext loss 0.18 · downstream score 0.60



Background-only features

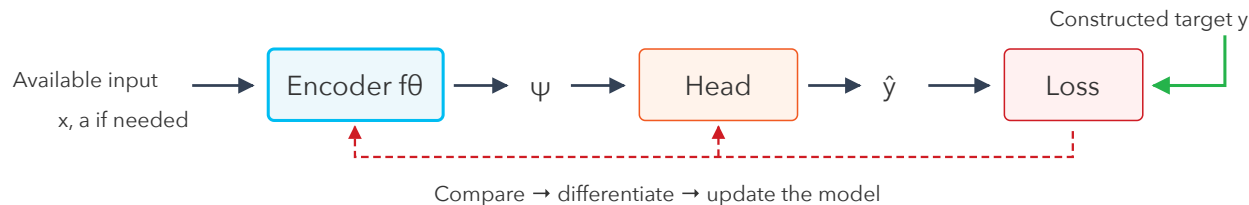
Views agree, but handle position is lost.

A lower pretext loss does not establish improvement on the intended robot task.

Course synthesis

Related views	Hidden observations	Teacher outputs	Geometry	Later measurements
$\tilde{x}_1 \leftrightarrow \tilde{x}_2$	$x_k \rightarrow \text{target } x_k$	$\text{sg}(f_{\theta}(x))$	Depth + poses	$o \rightarrow \Gamma \rightarrow y$
Augmentation / time	Withheld pixels	Target features	Depth / visible matches	Traversal / interaction

Alternative target sources, each with a defined construction rule and supported observations.



What the signal assumes View relations · representation · visibility · timing · measurement support

Reuse the representation, or use the learned task predictor directly.
Evaluate its output on the intended held-out robot task.

COURSE SOURCES

References



References: representation learning

- | | |
|---------------|---|
| SimCLR | A Simple Framework for Contrastive Learning of Visual Representations
T. Chen, S. Kornblith, M. Norouzi, G. Hinton. ICML 2020 . |
| DINO | Emerging Properties in Self-Supervised Vision Transformers
M. Caron, H. Touvron, I. Misra, H. Jégou, J. Mairal, P. Bojanowski, A. Joulin. ICCV 2021 . |
| VICReg | VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning
A. Bardes, J. Ponce, Y. LeCun. ICLR 2022 . |
| MAE | Masked Autoencoders Are Scalable Vision Learners
K. He, X. Chen, S. Xie, Y. Li, P. Dollár, R. Girshick. CVPR 2022 . |
| I-JEPA | Self-Supervised Learning From Images With a Joint-Embedding Predictive Architecture
M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, N. Ballas. CVPR 2023 .
Official training code: src/train.py |

Other sources cited in the course

- | | |
|-----------------|--|
| FixMatch | FixMatch: Simplifying Semi-Supervised Learning with Consistency and Confidence
K. Sohn, D. Berthelot, C.-L. Li, Z. Zhang, N. Carlini, E. D. Cubuk, A. Kurakin, H. Zhang, C. Raffel. NeurIPS 2020 . |
| NetVLAD | NetVLAD: CNN Architecture for Weakly Supervised Place Recognition
R. Arandjelović, P. Gronat, A. Torii, T. Pajdla, J. Sivic. CVPR 2016 . |

References: geometry and correspondence

Monodepth2

Digging Into Self-Supervised Monocular Depth Estimation

C. Godard, O. Mac Aodha, M. Firman, G. Brostow. **ICCV 2019.**

Official depth evaluator: [evaluate_depth.py](#) · commit b017a81

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Time-Contrastive Networks: Self-Supervised Learning from Video

P. Sermanet, C. Lynch, Y. Chebotar, J. Hsu, E. Jang, S. Schaal, S. Levine. **ICRA 2018.**

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Dense Object Nets: Learning Dense Visual Object Descriptors By and For Robotic Manipulation

P. R. Florence, L. Manuelli, R. Tedrake. **CoRL 2018.**

References: interaction and evaluation

WVN

Wild visual navigation: fast traversability learning via pre-trained models and online self-supervision

M. Mattamala, J. Frey, P. Libera, N. Chebrolu, G. Martius, C. Cadena, M. Hutter, M. Fallon.

Autonomous Robots 49, article 19, 2025.

ActAIM

Self-Supervised Learning of Action Affordances as Interaction Modes

L. Wang, N. Dvornik, R. Dubeau, M. Mittal, A. Garg. **ICRA 2023.**

Evaluation source

Evaluation

Cross-validation: evaluating estimator performance – grouped-data iterators

scikit-learn developers. **Official documentation • accessed 7 October 2026.**